

# LIMITS OF MATRIX APPROXIMATION OF GROUPS IN OPERATOR NORM

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We show that not every countable group is MF, where MF means admitting finite-dimensional unitary models that are asymptotically multiplicative and asymptotically separate nonidentity elements in operator norm. More precisely, the simple property-(T) group  $H = \text{EL}_{12}(L_{\mathbb{F}_2}(1, 2))$  has trivial image under every homomorphism to an MF group; consequently its reduced group  $C^*$ -algebra is a separable stably finite  $C^*$ -algebra that is not MF. The obstruction comes from a one-sided compression phenomenon for Kazhdan subgroups in norm matrix coronas. In the opposite direction, we prove an MF permanence theorem for HNN extensions whose edge isomorphism is implemented by a unitary of a norm matrix corona. Combining the two mechanisms, we show that recognizing MF groups from finite presentations is  $\Pi_2^0$ -complete.

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CONSUMPTION**

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## 1. INTRODUCTION

Not every countable group is MF. Our counterexample is

$$H = \text{EL}_{12}(L_{\mathbb{F}_2}(1, 2)).$$

The group  $H$  is countable, nontrivial, simple, and has property (T), but every homomorphism from  $H$  to an MF group is trivial. In particular,  $H$  is not MF. Its reduced group  $C^*$ -algebra  $C_r^*(H)$  is separable and stably finite, but is not MF.

More precisely, a countable group  $G$  is *MF* if there are positive integers  $d_n$  and maps  $V_n: G \rightarrow \mathcal{U}(d_n)$ , with  $V_n(1) = 1$ , such that

$$\|V_n(gh) - V_n(g)V_n(h)\| \rightarrow 0 \quad (g, h \in G),$$

and

$$\limsup_n \|V_n(g) - 1\| > 0 \quad (g \in G \setminus \{1\}).$$

This is the operator norm notion of MF group introduced by Carrión–Dadarlat–Eckhardt [CDE].

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Equivalently,  $G$  embeds in the unitary group of the norm matrix corona associated to the sequence  $\mathbf{d} = (d_n)$ ,

$$\mathcal{Q}_{\mathbf{d}} = \prod_n M_{d_n}(\mathbb{C}) / \bigoplus_n M_{d_n}(\mathbb{C}).$$

Here

$$\bigoplus_n M_{d_n}(\mathbb{C}) = \{(x_n) \in \prod_n M_{d_n}(\mathbb{C}) : \|x_n\| \rightarrow 0\}$$

is the  $c_0$ -direct sum.

A separable  $C^*$ -algebra is called MF if it embeds as a  $C^*$ -subalgebra of a norm matrix corona. For every countable group  $G$ , the reduced group algebra  $C_r^*(G)$  is separable and has a faithful canonical trace, hence is stably finite. Suppose that  $e: C_r^*(G) \rightarrow \mathcal{Q}_{\mathbf{d}}$  is an MF embedding and put  $p = e(1)$ . Then  $g \mapsto e(\lambda_g) + (1 - p)$  is an injective homomorphism from  $G$  into  $\mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ , so  $G$  is MF. Therefore a non-MF group automatically gives a separable stably finite reduced group  $C^*$ -algebra that is not MF.

For a countable group  $G$ , define its MF radical by

$$\text{Rad}_{\text{MF}}(G) = \bigcap_{\mathbf{d}} \bigcap_{\pi: G \rightarrow \mathcal{U}(\mathcal{Q}_{\mathbf{d}})} \ker \pi.$$

Proposition 2.1 shows that  $G$  is MF precisely when  $\text{Rad}_{\text{MF}}(G)$  is trivial. If  $\text{Rad}_{\text{MF}}(G) = G$ , every homomorphism from  $G$  to an MF group is trivial.

For a subgroup  $L \leq G$ , put

$$\text{Comp}_G(L) = \{u \in G : uLu^{-1} \leq L\}$$

and define its *compression-centralizer defect* by

$$(1) \quad \mathfrak{D}_G(L) = \langle\langle [ucu^{-1}, \ell] : u \in \text{Comp}_G(L), c \in C_G(L), \ell \in L \rangle\rangle_G.$$

**Theorem A** (one-sided compression criterion). *Let  $G$  be countable and let  $L \leq G$  have property (T). If  $K \trianglelefteq G$  has property (T) and  $K \leq \mathfrak{D}_G(L)$ , then*

$$K \leq \text{Rad}_{\text{MF}}(G).$$

*In particular, a nontrivial such  $K$  makes  $G$  non-MF. If  $G$  has property (T) and  $\mathfrak{D}_G(L) = G$ , then  $\text{Rad}_{\text{MF}}(G) = G$ .*

We sketch the argument. Corollary 2.6 gives  $\mathfrak{D}_G(L) \leq R_{\infty \rightarrow 2}(G)$ . Equivalently, if  $(V_n)$  is any operator norm asymptotic representation of  $G$ , then  $\|V_n(d) - 1\|_2 \rightarrow 0$  for every  $d \in \mathfrak{D}_G(L)$ . To upgrade this conclusion to corona triviality on  $K$ , fix a homomorphism  $\Theta: G \rightarrow \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$  and suppose, toward a contradiction, that  $\Theta$  is nontrivial on  $K$ . Let  $p$  be the image in  $\mathcal{Q}_{\mathbf{d}}$  of the Kazhdan projection of  $K$ . Because  $K \trianglelefteq G$ , the projection  $p$  commutes with  $\Theta(G)$ ; hence  $q = 1 - p$  defines a  $G$ -invariant complementary corner. After restricting coordinates and identifying each nonzero corner with a full matrix algebra, Lemma 2.7 realizes this action by an operator norm asymptotic representation  $W_n: G \rightarrow \mathcal{U}(r_n)$ . Its restriction to  $K$  has no nonzero fixed vectors, so the Kazhdan inequality yields a single  $s_0 \in K$  and a subsequence along which  $\|W_n(s_0) - I_{r_n}\|_2$  stays bounded below by a positive constant.

This contradicts  $s_0 \in K \leq \mathfrak{D}_G(L)$  and the preceding convergence. Thus every corona homomorphism kills  $K$ , as required.

Theorem 2.2 gives a separate finite-dimensional vanishing result: every finite-dimensional linear representation of  $G$  over any field kills  $\mathfrak{D}_G(L)$ .

We next apply Theorem A to the binary Leavitt construction that produces the group  $H$  announced above.

Let  $L_{\mathbb{F}_2}(1, 2)$  denote the binary Leavitt algebra. It is the unital  $\mathbb{F}_2$ -algebra generated by  $s_0, s_1, t_0, t_1$  subject to

$$(2) \quad t_i s_j = \delta_{ij}, \quad s_0 t_0 + s_1 t_1 = 1.$$

For a unital ring  $R$ , write  $e_{ij}(a) = 1 + aE_{ij}$  and let  $\text{EL}_n(R)$  be the subgroup of  $\text{GL}_n(R)$  generated by the elements  $e_{ij}(a)$ ,  $i \neq j$ .

**Theorem B** (the binary Leavitt group). *Put  $R = L_{\mathbb{F}_2}(1, 2)$  and  $H = \text{EL}_{12}(R)$ . Then  $H$  is countable, nontrivial, simple, has property (T), and*

$$\text{Rad}_{\text{MF}}(H) = H.$$

*Equivalently, every homomorphism from  $H$  to an MF group is trivial. In particular,  $H$  is non-MF. Its reduced group  $C^*$ -algebra  $C_r^*(H)$  is separable and stably finite, but is not MF.*

Inside  $H$ , an explicit element  $\tau$  satisfies  $\tau L \tau^{-1} \leq L$  for the subgroup  $L \cong \text{EL}_3(R)$  in the upper left. The element  $c = e_{34}(1)$  centralizes  $L$ , and for  $\ell = e_{12}(1)$  the corresponding commutator is

$$d = e_{02}(s_1 t_1).$$

The relation  $t_1(s_1 t_1)s_1 = 1$  shows that  $d \neq 1$ . Since  $H$  is simple,  $d$  normally generates  $H$ . Since  $d \in \mathfrak{D}_H(L)$  and  $\mathfrak{D}_H(L)$  is normal, it follows that  $\mathfrak{D}_H(L) = H$ . Therefore Theorem A gives  $\text{Rad}_{\text{MF}}(H) = H$ .

If  $\rho: G \rightarrow \text{GL}(V)$  is a homomorphism and  $V$  is finite-dimensional, conjugation by  $\rho(u)^{-1}$  maps the commutant of  $\rho(L)$  injectively into itself, hence bijectively. Thus  $\rho$  kills  $\mathfrak{D}_G(L)$ . Consequently,  $\mathfrak{D}_G(L) = 1$  if  $G$  has a faithful finite-dimensional linear representation. The same conclusion holds if  $G$  is residually finite, because every nonidentity element then survives in a finite permutation representation. If  $G$  is amenable, its property-(T) subgroup  $L$  is finite [BHV]; then  $uLu^{-1} = L$ , so  $ucu^{-1}$  centralizes  $L$  for every  $u \in \text{Comp}_G(L)$ , and again  $\mathfrak{D}_G(L) = 1$ .

The argument is specific to operator norm approximation and gives no conclusion about soficity or hyperlinearity: normalized Hilbert–Schmidt approximations do not provide the operator norm control used to construct the conjugation representation in Theorem 2.5.

The final main result concerns all finitely presented groups at once. It locates the recognition problem for the MF property in the Kleene–Mostowski arithmetical hierarchy, whose levels  $\Sigma_n^0$  and  $\Pi_n^0$  count alternations of quantifiers over a decidable relation.

**Theorem C** (recognizing MF groups). *Fix an effective coding of finite presentations of groups by natural numbers. The set of codes of finite presentations of MF groups is  $\Pi_2^0$ -complete, and the set of codes of finite presentations of non-MF groups is  $\Sigma_2^0$ -complete. In particular, neither set is recursively enumerable, and no algorithm decides from a finite presentation whether the presented group is MF. More precisely, there is a computable map  $e \mapsto \hat{R}_e$  into finite presentations such that the group presented by  $\hat{R}_e$  is MF if and only if the  $e$ -th partial computable function has infinite domain.*

The upper bound is a certificate normal form: the MF property is determined by finite data with a fixed separation constant, and the existence of a certificate at a given scale is decidable (Section 4). For the lower bound we encode the  $\Pi_2^0$ -complete set of programs with infinite domain. The construction of Bilanovic–Chubb–Roven gives a recursively presented group that is trivial or contains  $H$  according as the domain is infinite or finite; an effective Higman embedding and two HNN extensions turn it into a finitely presented group (Section 6); and Theorem 5.2 proves that the resulting group is MF when the domain is infinite: an HNN extension of a group with a tracial MF realization is MF whenever the edge isomorphism is implemented by a unitary of a norm matrix corona (Sections 5 and 7).

Theorems A and 5.2 point in opposite directions. The first turns a one-sided conjugation in a norm matrix corona into an obstruction to operator norm approximation; the second turns a conjugation implemented in a norm matrix corona into a construction that preserves it. Theorem C comes from building a finite presentation in which one or the other mechanism wins according to a  $\Pi_2^0$ -complete predicate: on one branch the group contains  $H$ , on the other it falls under Theorem 5.2. Finite presentability by itself is not the point. The MF property is closed in the space of marked groups (Lemma 4.2), so every finitely generated non-MF group is a quotient of a finitely presented one; the content of Theorem C is the uniform computable family and the exact level. Section 8 records a complementary calculus that the main line does not use: the MF radical pulls back along surjections whose kernel is invisible to MF, and every countable group  $Q$  is the quotient visible to MF of a non-MF group  $W_Q$ .

**Relation to prior work.** Blackadar and Kirchberg introduced the notions of MF and NF algebras in a 1994 Oberwolfach abstract [BK94]. They wrote that the classes of NF, strong NF, and separable nuclear stably finite  $C^*$ -algebras might coincide. Their 1997 paper developed these notions and proved that a separable  $C^*$ -algebra is NF if and only if it is nuclear and MF [BK]. It also asked whether every separable nuclear stably finite  $C^*$ -algebra is quasidiagonal. This was a question about nuclear  $C^*$ -algebras, not about arbitrary groups.

The group property used here was introduced by Carrión–Dadarlat–Eckhardt [CDE]. If every separable stably finite  $C^*$ -algebra were MF, then every countable group would be MF because its reduced group  $C^*$ -algebra is separable

and stably finite. Later approximation papers described the resulting operator norm group problem as related to Kirchberg’s question [LO, Th]. The negative solution of the Connes embedding problem [JNVWY] yields separable stably finite  $C^*$ -algebras that are not MF, as Goldbring and Hart observe [GH, Proposition 6.1 and Remark 6.2]. Theorem B gives such an example among reduced group  $C^*$ -algebras. Korchagin subsequently developed permanence and local approximation results for MF groups [Kor]. Shulman proves that  $A *_C A$  is MF whenever  $A$  is a separable MF  $C^*$ -algebra and  $C \subseteq A$  is a  $C^*$ -subalgebra [Sh, Theorem 10]. She also proves that the full group  $C^*$ -algebras of amalgamated free products of amenable groups and of certain semidirect products are MF [Sh]. Nonapproximability was known for the unnormalized Schatten  $p$ -norms with  $1 < p < \infty$  [DGLT, LO]. Bachner–Dogan–Lubotzky later proved nonapproximability for  $p = 1$  [BDL]. The operator norm case  $p = \infty$  remained open. They also identify stability from operator norm to normalized Hilbert–Schmidt norm as a possible route to a non-MF group [BDL, Proposition 1.5]. Here operator norm control implies that coordinatewise conjugation defines a representation in a second norm matrix corona. The Kazhdan projection of this conjugation representation makes the Hilbert–Schmidt asymptotic commutant invariant under the one-sided compressor. Theorem 2.8 then proves that a normal property-(T) subgroup contained in the defect lies in the MF radical.

Leavitt introduced the algebras  $L_K(1, n)$  as examples of rings with nonunique module rank [Lev]. Their defining relations supply the internal compression used here. The binary Leavitt algebra is purely infinite simple [AA], hence an exchange ring [Ara], and its center is the base field [AC]. Preusser classified the subgroups of  $GL_n$  over an exchange ring that are normalized by  $EL_n$  [Pre]. His sandwich theorem, together with the simplicity of the binary Leavitt algebra and the equality  $Z(R) = \mathbb{F}_2$ , implies that  $EL_{12}(R)$  is simple (Proposition 3.3). Ershov–Jaikin–Zapirain give property (T) for  $EL_n(R)$  whenever  $n \geq 3$  and  $R$  is a finitely generated unital associative ring [EJZ, Theorem 1.1]. Related rigidity results for metric approximations of Kazhdan groups appear in [KT, AT, Dad].

Decision problems for finitely presented groups go back to Dehn, and the theorem of Adian and Rabin [Ad, Ra] shows that no Markov property can be recognized from a finite presentation. Once a finitely presented non-MF group exists, the MF property is Markov, and Adian–Rabin gives undecidability; Theorem C gives the exact level. Torsion-freeness is the only other property of finitely presented groups whose recognition problem was known to be  $\Pi_2^0$ -complete, by Lempp [Lem]. For recursively presented groups, Bilanovic–Chubb–Roven proved that every Markov property is  $\Pi_2^0$ -hard to recognize [BCR]; their construction is the first step of Section 6. The passage from recursive to finite presentations is where the operator algebra enters: a generic Higman embedding does not preserve the MF property, and the HNN extensions of Section 6 are chosen so that Theorem 5.2 applies to them.

## 2. ONE-SIDED COMPRESSION AND THE MF RADICAL

This section proves Theorem A. The mechanism is visible already in finite dimension: an injective linear endomorphism of a finite-dimensional space is surjective, so a one-sided invariant commutant is invariant. In a norm matrix corona the substitute for finite dimensionality is stable finiteness, which forces two equivalent projections, one below the other, to be equal; the projection in question is the Kazhdan projection of a property-(T) subgroup. Property (T) of a normal subgroup then transports the resulting Hilbert–Schmidt collapse back into an obstruction in operator norm.

**2.1. The MF radical.** For  $N \trianglelefteq G$  define its *MF kernel closure* by

$$\text{cl}_{\text{MF}}^G(N) = \bigcap \{\ker f : N \leq \ker f, f: G \rightarrow M, M \text{ MF}\}.$$

The image of a corona homomorphism from  $G$  is countable and is itself MF; conversely, every MF target embeds in a norm matrix corona. Thus  $\text{Rad}_{\text{MF}}(G) = \text{cl}_{\text{MF}}^G(1)$ , and  $G/N$  is MF precisely when  $\text{cl}_{\text{MF}}^G(N) = N$ . This is a closure operation on the normal subgroups of one fixed group; it is unrelated to topological closure of classes of marked groups.

**Proposition 2.1** (basic properties of the MF radical). *For every countable group  $G$ , the subgroup  $\text{Rad}_{\text{MF}}(G)$  is fully invariant, the quotient  $G/\text{Rad}_{\text{MF}}(G)$  is MF, and*

$$G/N \text{ is MF} \iff \text{cl}_{\text{MF}}^G(N) = N \quad (N \trianglelefteq G).$$

*In particular,  $G$  is MF if and only if its MF radical is trivial.*

*Proof.* An intersection of kernels is normal. If  $\alpha$  is an endomorphism of  $G$ ,  $x \in \text{Rad}_{\text{MF}}(G)$ , and  $\pi$  is a corona homomorphism from  $G$ , then  $\pi \circ \alpha$  kills  $x$ . Hence  $\pi(\alpha(x)) = 1$  for every  $\pi$ , which proves full invariance.

Put  $R = \text{Rad}_{\text{MF}}(G)$ . If  $G/R$  is trivial, there is nothing to prove. If  $x \in G/R$  is nontrivial and  $g \in G$  maps to  $x$ , then  $g \notin R$ . Some homomorphism from  $G$  to an MF group therefore separates  $g$  from the identity. It kills  $R$  and descends to a homomorphism from  $G/R$  that separates  $x$ . Thus  $G/R$  is residually MF, and hence MF by Korchagin [Kor, Proposition 6].

For  $N \trianglelefteq G$ , homomorphisms from  $G/N$  to MF groups correspond to homomorphisms from  $G$  to MF groups that kill  $N$ . It follows directly that

$$\text{Rad}_{\text{MF}}(G/N) = \text{cl}_{\text{MF}}^G(N)/N.$$

Since a countable group is MF exactly when its MF radical is trivial, this identity gives the asserted closure criterion.  $\square$

## 2.2. Finite dimension.

**Theorem 2.2** (finite-dimensional commutant rigidity). *Let  $G$  be a group, let  $L \leq G$ , and suppose  $u \in G$  satisfies  $uLu^{-1} \leq L$ . If  $k$  is a field,  $V$  is a finite-dimensional  $k$ -vector space, and  $\rho: G \rightarrow \text{GL}(V)$  is a homomorphism, then*

$$\rho(u)\rho(L)'\rho(u)^{-1} = \rho(L)',$$

where  $\rho(L)'$  is the commutant of  $\rho(L)$  in  $\text{End}_k(V)$ . Consequently, if  $c \in C_G(L)$ , then

$$\rho([ucu^{-1}, \ell]) = 1 \quad (\ell \in L).$$

*Proof.* Put  $C = \rho(L)'$ . If  $x \in C$ ,  $h \in L$ , and  $h' = uhu^{-1} \in L$ , then

$$\begin{aligned} \rho(h)\rho(u)^{-1}x\rho(u) &= \rho(u)^{-1}\rho(h')x\rho(u) \\ &= \rho(u)^{-1}x\rho(h')\rho(u) \\ &= \rho(u)^{-1}x\rho(u)\rho(h). \end{aligned}$$

Thus  $\rho(u)^{-1}C\rho(u) \subseteq C$ . Conjugation by  $\rho(u)^{-1}$  is injective, and  $C$  is finite-dimensional. Its restriction to  $C$  is therefore surjective, so the inclusion is equality. Conjugating by  $\rho(u)$  gives  $\rho(u)C\rho(u)^{-1} = C$ .

If  $c \in C_G(L)$ , then  $\rho(c) \in C$ , hence  $\rho(ucu^{-1}) \in C$ . Therefore  $\rho([ucu^{-1}, \ell]) = 1$  for every  $\ell \in L$ .  $\square$

This is where finite dimensionality enters the proof: an injective linear self-map of an infinite-dimensional commutant need not be surjective. In Section 2.3 the Kazhdan projection places the analogous endomorphism inside a norm matrix corona, where stable finiteness supplies the missing surjectivity for projections onto fixed vectors.

**2.3. Kazhdan transport in normalized Hilbert–Schmidt norm.** For  $a \in M_d(\mathbb{C})$  write

$$\|a\|_2 = \text{tr}_d(a^*a)^{1/2}, \quad \text{tr}_d = \frac{1}{d} \text{Tr}.$$

An *operator norm asymptotic representation* of a countable group  $G$  is a sequence of maps  $V_n: G \rightarrow \mathcal{U}(d_n)$  such that  $V_n(1) = 1$  and

$$\|V_n(gh) - V_n(g)V_n(h)\| \longrightarrow 0 \quad (g, h \in G).$$

For such a sequence put

$$K_2(V) = \{g \in G : \|V_n(g) - 1\|_2 \longrightarrow 0\}.$$

The operator norm defects also tend to zero in normalized Hilbert–Schmidt norm, so  $K_2(V)$  is a normal subgroup of  $G$ . Define

$$(3) \quad R_{\infty \rightarrow 2}(G) = \bigcap_V K_2(V),$$

where  $V$  ranges over all operator norm asymptotic representations of  $G$ .

For  $L \leq G$ , let

$$\mathcal{C}_2(V, L) = \left\{ (x_n) \left| \begin{array}{l} \sup_n \|x_n\| < \infty, \\ \|[V_n(\ell), x_n]\|_2 \longrightarrow 0 \quad (\ell \in L) \end{array} \right. \right\}$$

be the bounded Hilbert–Schmidt asymptotic commutant of  $V(L)$ . Coordinatewise conjugation by  $V_n(g)$  defines a bijection, denoted  $\text{Ad}(V(g))$ , of the bounded matrix sequences.

**Lemma 2.3.** *For every sequence of positive integers  $(m_n)$ , the norm matrix corona*

$$\prod_n M_{m_n}(\mathbb{C}) / \bigoplus_n M_{m_n}(\mathbb{C})$$

*is stably finite. Consequently, unitarily equivalent projections in a norm matrix corona are equal whenever one is dominated by the other.*

*Proof.* Suppose that  $v^*v = 1$  in the corona, and let  $(x_n)$  be a bounded lift of  $v$ . Then  $x_n^*x_n \rightarrow 1$  in norm. For all sufficiently large  $n$ , the matrix  $x_n$  is invertible, and the unitary  $x_n(x_n^*x_n)^{-1/2}$  differs from  $x_n$  by  $o(1)$ . Thus  $v$  is represented by unitaries and  $vv^* = 1$ . The same argument applies to every matrix amplification of the corona. If  $A$  is a stably finite unital  $C^*$ -algebra, equivalent projections  $p \leq q$  satisfy  $p = q$ . Indeed, if  $w^*w = q$  and  $ww^* = p$ , then  $w \in qAq$  is an isometry in the corner. The corner is finite because  $w + (1 - q)$  is an isometry in  $A$ . Consequently  $ww^* = q$ , and hence  $p = q$ .  $\square$

**Lemma 2.4** (one-sided order for the Kazhdan projection). *Let  $L$  have property (T), let  $B$  be a unital  $C^*$ -algebra, and let  $\pi: L \rightarrow \mathcal{U}(B)$  be a homomorphism. Denote by  $P \in B$  the image of the Kazhdan projection under the extension  $C_{\max}^*(L) \rightarrow B$ . If  $U \in \mathcal{U}(B)$  satisfies*

$$U\pi(L)U^* \subseteq \pi(L),$$

*then*

$$U^*PU \leq P.$$

*Proof.* Represent  $B$  faithfully and nondegenerately on a Hilbert space  $\mathcal{H}$ . The represented projection  $P$  is the orthogonal projection onto  $\text{Fix } \pi(L)$ . If  $\xi$  is  $L$ -fixed and  $\ell \in L$ , then

$$\pi(\ell)U^*\xi = U^*(U\pi(\ell)U^*)\xi = U^*\xi,$$

because  $U\pi(\ell)U^*$  belongs to  $\pi(L)$ . Hence  $U^*\text{Fix } \pi(L) \subseteq \text{Fix } \pi(L)$ , so the range projection  $U^*PU$  is dominated by  $P$  in  $B(\mathcal{H})$ . Faithfulness of the representation gives the same projection inequality in  $B$ .  $\square$

**Theorem 2.5** (one-sided Kazhdan transport). *Let  $L \leq G$  have property (T), and let  $u \in G$  satisfy  $uLu^{-1} \leq L$ . Let  $(V_n)$  be an operator norm asymptotic representation of  $G$ . Then*

$$\text{Ad}(V(u))(\mathcal{C}_2(V, L)) = \mathcal{C}_2(V, L).$$

*Proof.* Fix  $x = (x_n) \in \mathcal{C}_2(V, L)$  and let  $\omega$  be a free ultrafilter. Regard  $M_{d_n}(\mathbb{C})$  as a Hilbert space with its normalized Hilbert–Schmidt inner product, and form the ultraproduct of Hilbert spaces  $\mathcal{K}_\omega$ . Conjugation by the  $V_n(g)$  defines a unitary representation  $\sigma$  of  $G$  on  $\mathcal{K}_\omega$ . Indeed, for unitaries  $A, B$  one has

$$\|\text{Ad}(A) - \text{Ad}(B)\| \leq 2\|A - B\|,$$

so the operator norm multiplicative defects of  $(V_n)$  also make the conjugation actions multiplicative modulo  $c_0$ . The class  $\xi = [x_n]_\omega$  is fixed by  $\sigma(L)$ .



For  $g \in G$ , let  $\tilde{\sigma}(g)$  be the class of the coordinate operators  $\text{Ad}(V_n(g))$  in the norm matrix corona

$$\mathcal{B} = \prod_n B(M_{d_n}(\mathbb{C})) / \bigoplus_n B(M_{d_n}(\mathbb{C})).$$

After choosing matrix units on each  $M_{d_n}(\mathbb{C})$ , this is a norm matrix corona with coordinate sizes  $d_n^2$ . The preceding estimate shows directly that  $g \mapsto \tilde{\sigma}(g)$  is an exact homomorphism  $G \rightarrow \mathcal{U}(\mathcal{B})$ . Its restriction to  $L$  therefore extends to a  $*$ -homomorphism  $C_{\max}^*(L) \rightarrow \mathcal{B}$ . Let  $P \in \mathcal{B}$  be the image of the Kazhdan projection [AW]. Under the natural representation of  $\mathcal{B}$  on  $\mathcal{K}_\omega$ , its range is  $\text{Fix } \sigma(L)$ . This representation of  $\mathcal{B}$  need not be faithful; it is used only after the following projection identity has been proved inside  $\mathcal{B}$ . Since  $uLu^{-1} \leq L$ , the unitary  $U = \tilde{\sigma}(u)$  satisfies the hypothesis of Lemma 2.4, and hence  $U^*PU \leq P$  inside  $\mathcal{B}$ . The two projections are unitarily equivalent. Lemma 2.3 therefore gives  $U^*PU = P$ , so  $U$  commutes with  $P$ . Hence both  $U\xi$  and  $U^*\xi$  are fixed by  $\sigma(L)$ . Since  $\omega$  was arbitrary, the corresponding Hilbert–Schmidt commutators converge to zero along the full sequence. Thus  $\text{Ad}(V(u))$  and its inverse both preserve  $\mathcal{C}_2(V, L)$ , which proves the equality.  $\square$

**Corollary 2.6.** *Let  $L \leq G$  have property (T), let  $uLu^{-1} \leq L$ , and let  $c \in C_G(L)$ . Then*

$$[ucu^{-1}, \ell] \in R_{\infty \rightarrow 2}(G) \quad (\ell \in L).$$

*Proof.* Let  $(V_n)$  be an operator norm asymptotic representation. If  $c \in C_G(L)$ , then  $(V_n(c)) \in \mathcal{C}_2(V, L)$ . By Theorem 2.5, the same is true of  $(V_n(u)V_n(c)V_n(u)^*)$ . Asymptotic multiplicativity identifies this sequence in operator norm with  $(V_n(ucu^{-1}))$ ; explicitly,

$$\|V_n(u)V_n(c)V_n(u)^* - V_n(ucu^{-1})\| \longrightarrow 0.$$

Therefore

$$\|V_n([ucu^{-1}, \ell]) - 1\|_2 \longrightarrow 0 \quad (\ell \in L).$$

Intersecting over  $(V_n)$  proves the claim.  $\square$

**2.4. Normal Kazhdan subgroups and the MF radical.** Let  $K \trianglelefteq G$  have property (T), and let  $p$  be the image of its Kazhdan projection under a corona representation of  $G$ . Normality of  $K$  implies that  $p$  commutes with the image of  $G$ . Hence  $q = 1 - p$  also commutes with that image, and in every nondegenerate representation of the corner  $q\mathcal{Q}_{\mathbf{d}}q$  the induced representation of  $K$  has no nonzero fixed vectors.

**Lemma 2.7** (central corona corners). *Let  $\rho: G \rightarrow \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$  be a homomorphism from a countable group, and let  $q \in \mathcal{Q}_{\mathbf{d}}$  be a nonzero projection commuting with  $\rho(G)$ . Then, after passing to an infinite coordinate subsequence, there are nonzero projections  $q_n \in M_{d_n}(\mathbb{C})$ , integers  $r_n = \text{rank}(q_n)$ ,*

unitary identifications  $J_n: \mathbb{C}^{r_n} \rightarrow q_n \mathbb{C}^{d_n}$ , and an operator norm asymptotic representation

$$W_n: G \longrightarrow \mathcal{U}(r_n)$$

with  $W_n(1) = I_{r_n}$ . In the corona over the retained coordinates, the class of  $(J_n W_n(g) J_n^*)$  is the coordinate restriction of  $q\rho(g)$ .

*Proof.* Lift  $q$  to projections  $q_n$ , and lift each  $\rho(g)$  to unitaries  $U_n(g)$ , choosing  $U_n(1) = I_{d_n}$ . The first lift is obtained by spectral rounding; the second is obtained by polar correction, since the two unitarity defects of any lift converge to zero in operator norm. The commutation relation in the corona gives

$$\|q_n U_n(g) - U_n(g) q_n\| \longrightarrow 0 \quad (g \in G).$$

Consequently  $q_n U_n(g) q_n$ , viewed on  $q_n \mathbb{C}^{d_n}$ , has both unitarity defects converging to zero. For each fixed  $g$ , let  $\widetilde{W}_n(g)$  be its polar correction whenever both defects are at most  $1/2$ , and set  $\widetilde{W}_n(g) = q_n$  at the finitely many earlier stages. Then  $\widetilde{W}_n(g)$  is unitary in the corner and satisfies

$$\|\widetilde{W}_n(g) - q_n U_n(g) q_n\| \longrightarrow 0.$$

For  $g = 1$ , the compression is already the identity  $q_n$  of the corner, so take  $\widetilde{W}_n(1) = q_n$ . Since  $q \neq 0$ , infinitely many  $q_n$  are nonzero. Retain and relabel those coordinates. Put  $r_n = \text{rank}(q_n)$ , choose a unitary  $J_n: \mathbb{C}^{r_n} \rightarrow q_n \mathbb{C}^{d_n}$ , and define

$$W_n(g) = J_n^* \widetilde{W}_n(g) J_n \in \mathcal{U}(r_n).$$

Conjugation by  $J_n$  preserves operator norms and multiplicative defects. Thus  $(W_n)$  is an operator norm asymptotic representation with  $W_n(1) = I_{r_n}$ . The displayed estimate identifies the class of  $(J_n W_n(g) J_n^*)$  with the coordinate restriction of  $q\rho(g)$ .  $\square$

**Theorem 2.8** (normal Kazhdan radical theorem). *Let  $G$  be countable and let  $K \trianglelefteq G$  have property (T) with  $K \leq R_{\infty \rightarrow 2}(G)$ . Then*

$$K \leq \text{Rad}_{\text{MF}}(G).$$

*Proof.* Suppose that a homomorphism  $\Theta: G \rightarrow \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$  is nontrivial on  $K$ . Replace  $G$  by  $\Theta(G)$ , replace  $K$  by  $\Theta(K)$ , and denote the resulting inclusion into the corona again by  $\Theta$ . The new  $K$  is a nontrivial normal property-(T) subgroup, since property (T) passes to quotients. Every one of its elements remains in the universal Hilbert–Schmidt kernel, because any operator norm asymptotic representation of  $\Theta(G)$ , precomposed with the quotient map from the original group, is such a representation of the original group.

Let  $p_K \in C_{\max}^*(K)$  be the Kazhdan projection and let  $p$  be its image in  $\mathcal{Q}_{\mathbf{d}}$ . Under any faithful representation of the corona,  $p$  is the projection onto the  $K$ -fixed vectors. Normality of  $K$  gives

$$\Theta(g) p \Theta(g)^* = p \quad (g \in G).$$

Indeed, in a faithful representation of the corona the two projections have ranges  $\Theta(g) \text{Fix } \Theta(K)$  and  $\text{Fix } \Theta(K)$ ; these agree because  $gKg^{-1} = K$ . Put  $q = 1 - p$ . The projection  $q$  is nonzero: if  $p = 1$ , then every vector is fixed by  $K$ , so every element of  $K$  has trivial corona image. Lemma 2.7 gives positive integers  $r_n$  and an operator norm asymptotic representation  $W_n: G \rightarrow \mathcal{U}(r_n)$ . Let  $\hat{\Theta}: G \rightarrow \mathcal{U}(\mathcal{Q}_r)$  be its corona homomorphism. The coordinatewise corner identifications carry  $\hat{\Theta}$  to the coordinate restriction of  $g \mapsto q\Theta(g)$ .

Choose a finite Kazhdan set  $S \subseteq K$  and a Kazhdan constant  $\kappa > 0$ . The Kazhdan projection of  $K$  under  $\hat{\Theta}$  is zero. Hence the positive element

$$b = \frac{1}{|S|} \sum_{s \in S} (\hat{\Theta}(s) - 1)^* (\hat{\Theta}(s) - 1)$$

satisfies

$$b \geq \frac{\kappa^2}{|S|} 1.$$

Represent  $\mathcal{Q}_r$  faithfully. Because the image of the Kazhdan projection is zero,  $\hat{\Theta}|_K$  has no nonzero fixed vectors. The defining Kazhdan inequality therefore gives this operator inequality.

The coordinate elements

$$b_n = \frac{1}{|S|} \sum_{s \in S} (W_n(s) - I_{r_n})^* (W_n(s) - I_{r_n})$$

represent  $b$ . Taking the normalized trace on  $M_{r_n}(\mathbb{C})$  yields

$$\frac{1}{|S|} \sum_{s \in S} \|W_n(s) - I_{r_n}\|_2^2 \geq \frac{\kappa^2}{|S|} - o(1).$$

The order inequality for  $b$  says that the negative part of  $b_n - (\kappa^2/|S|)I_{r_n}$  converges to zero in operator norm. Taking normalized traces gives the displayed inequality. After passing to a subsequence, one fixed  $s_0 \in S$  therefore stays a positive Hilbert–Schmidt distance from  $I_{r_n}$ . This contradicts  $s_0 \in K \leq R_{\infty \rightarrow 2}(G)$ , established after passing to  $\Theta(G)$ . Thus every corona homomorphism kills  $K$ , as required.  $\square$

*Proof of Theorem A.* For every  $u \in \text{Comp}_G(L)$ ,  $c \in C_G(L)$ , and  $\ell \in L$ , Corollary 2.6 gives

$$[ucu^{-1}, \ell] \in R_{\infty \rightarrow 2}(G).$$

Normality of  $R_{\infty \rightarrow 2}(G)$  therefore gives

$$\mathfrak{D}_G(L) \leq R_{\infty \rightarrow 2}(G).$$

Since  $K \leq \mathfrak{D}_G(L)$ , Theorem 2.8 now gives  $K \leq \text{Rad}_{\text{MF}}(G)$ . Taking a nontrivial  $K$  gives the non-MF conclusion. Taking  $K = G = \mathfrak{D}_G(L)$  gives  $\text{Rad}_{\text{MF}}(G) = G$ .

For a finite-dimensional linear representation over an arbitrary field, Theorem 2.2 kills every generator of  $\mathfrak{D}_G(L)$ . Its kernel is normal, so it contains  $\mathfrak{D}_G(L)$ .  $\square$

### 3. THE BINARY LEAVITT GROUP

This section proves Theorem B. The defining relation  $R \cong R^2$  of the binary Leavitt algebra supplies the one-sided self-compression required by Theorem A; simplicity of  $\text{EL}_{12}(R)$  then turns a single invisible commutator into a full MF radical.

**3.1. The self-compression.** Throughout this section  $R = L_{\mathbb{F}_2}(1, 2)$  and

$$p = s_0 t_0, \quad q = s_1 t_1.$$

The relations (2) give

$$(4) \quad p + q = 1, \quad t_1 q s_1 = 1.$$

In particular  $q \neq 0$ .

We use indices  $0, \dots, 11$  for  $12 \times 12$  matrices. We identify  $\text{GL}_3(R)$  with the subgroup of  $\text{GL}_{12}(R)$  consisting of the matrices  $\text{diag}(A, I_9)$ . Define a map  $\Psi: M_3(R) \rightarrow M_3(R)$  by

$$(5) \quad \Psi(A) = qI_3 + s_0 A t_0.$$

Here  $s_0 A t_0$  denotes the matrix  $(s_0 a_{ij} t_0)_{ij}$ . The relations  $q^2 = q$ ,  $q s_0 = t_0 q = 0$ , and  $t_0 s_0 = 1$ , all consequences of (2), show that  $\Psi$  is multiplicative. Moreover,

$$\Psi(I_3) = qI_3 + s_0 I_3 t_0 = qI_3 + pI_3 = (p + q)I_3 = I_3,$$

so  $\Psi$  preserves the identity. Finally,  $t_0 \Psi(A) s_0 = A$ , and hence  $\Psi$  is injective. These identities show that  $\Psi$  is injective, multiplicative, and identity preserving, so its restriction to  $\text{GL}_3(R)$  is an injective group homomorphism  $\text{GL}_3(R) \rightarrow \text{GL}_3(R)$ .

To realize  $\Psi$  by conjugation on the embedded copy of  $\text{GL}_3(R)$ , define the following  $6 \times 6$  block matrices:

$$X = \begin{pmatrix} s_0 I_3 & s_1 t_0 I_3 \\ 0 & t_1 I_3 \end{pmatrix}, \quad Y = \begin{pmatrix} t_0 I_3 & 0 \\ s_0 t_1 I_3 & s_1 I_3 \end{pmatrix}.$$

A block multiplication using (2) gives  $XY = YX = I_6$ , so  $Y = X^{-1}$ . Put

$$(6) \quad \tau = \text{diag}(X, Y) \in \text{GL}_{12}(R).$$

A direct block multiplication gives

$$(7) \quad \tau \text{diag}(A, I_9) \tau^{-1} = \text{diag}(\Psi(A), I_9).$$

**Lemma 3.1.** *The matrix  $\tau$  defined in (6) belongs to  $\text{EL}_{12}(R)$ .*

*Proof.* Put  $I = I_6$ . The Whitehead factorization gives

$$(8) \quad \text{diag}(X, X^{-1}) = \begin{pmatrix} I & X \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ -X^{-1} & I \end{pmatrix} \begin{pmatrix} I & X \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ I & I \end{pmatrix} \begin{pmatrix} I & -I \\ 0 & I \end{pmatrix} \begin{pmatrix} I & 0 \\ I & I \end{pmatrix}.$$

If  $B = (b_{rs})_{r,s=0}^5 \in M_6(R)$ , then

$$\begin{pmatrix} I & B \\ 0 & I \end{pmatrix} = \prod_{r,s=0}^5 e_{r,6+s}(b_{rs}), \quad \begin{pmatrix} I & 0 \\ B & I \end{pmatrix} = \prod_{r,s=0}^5 e_{6+r,s}(b_{rs}).$$

In either product, distinct off-diagonal matrix units have product zero, so no cross-terms occur. Every factor on the right-hand side of (8) therefore belongs to  $\text{EL}_{12}(R)$ . Since  $Y = X^{-1}$ , its left-hand side is  $\tau$ , and the result follows.  $\square$

Set  $H = \text{EL}_{12}(R)$ , and let  $L = \text{EL}_3(R) \leq H$  denote the image of the embedding  $A \mapsto \text{diag}(A, I_9)$ .

**Proposition 3.2.** *The groups  $H$  and  $L$  have property (T), and  $\tau \in H$  satisfies*

$$(9) \quad \tau L \tau^{-1} \leq L.$$

*Proof.* The ring  $R$  is generated by  $s_0, s_1, t_0, t_1$  as a unital ring. The theorem of Ershov and Jaikin-Zapirain therefore gives property (T) for  $\text{EL}_{12}(R)$  and  $\text{EL}_3(R)$  [EJZ, Theorem 1.1]. Lemma 3.1 gives  $\tau \in H$ .

For  $i \neq j$  and  $a \in R$ , equations (7) and (5) give

$$\tau e_{ij}(a) \tau^{-1} = e_{ij}(s_0 a t_0) \in L,$$

where the elementary matrices are viewed in the embedded copy of  $L$ . These matrices generate  $L$ , so (9) follows.  $\square$

### 3.2. Simplicity and the full MF radical.

**Proposition 3.3.** *The group  $H = \text{EL}_{12}(R)$  is nontrivial and simple.*

*Proof.* The ring  $R$  is purely infinite simple [AA], and therefore is an exchange ring [Ara]. Let  $N \trianglelefteq H$ . Since  $H = \text{EL}_{12}(R)$ , the subgroup  $N \leq \text{GL}_{12}(R)$  is normalized by  $\text{EL}_{12}(R)$ . Preusser's sandwich theorem [Pre, Theorem 3] provides an ideal  $I \trianglelefteq R$  such that

$$\text{EL}_{12}(R, I) \leq N \leq C_{12}(R, I).$$

Simplicity of  $R$  gives  $I = 0$  or  $I = R$ . If  $I = R$ , then  $\text{EL}_{12}(R, I) = H$ , so  $N = H$ .

Suppose that  $I = 0$ . Then  $N \leq C_{12}(R, 0) = Z(\text{GL}_{12}(R))$ . A matrix that commutes with every  $e_{ij}(1)$  is a scalar matrix  $\lambda I_{12}$ . Commutation with  $e_{ij}(a)$  for arbitrary  $a \in R$  then gives  $\lambda a = a \lambda$ , so  $\lambda \in Z(R)^\times$ . Since  $Z(R) = \mathbb{F}_2$  [AC, Corollary 4.3], we have  $\lambda = 1$  and  $N = 1$ . Finally,  $e_{01}(1) \neq 1$ , so  $H$  is nontrivial.  $\square$

**Proposition 3.4.** *Let*

$$c = e_{34}(1), \quad \ell = e_{12}(1), \quad d = [\tau c \tau^{-1}, \ell]$$

*in  $H$ . Then*

$$c \in C_H(L), \quad d = e_{02}(q) \neq 1, \quad \langle\langle d \rangle\rangle_H = H.$$

*Proof.* For  $i, j \in \{0, 1, 2\}$  with  $i \neq j$  and  $a \in R$ , we have  $j \neq 3$  and  $i \neq 4$ . Hence the Steinberg relations give

$$[e_{ij}(a), e_{34}(1)] = 1.$$

These matrices generate  $L$ , so  $c = e_{34}(1)$  belongs to  $C_H(L)$ .

Direct multiplication of the matrices in (6) gives

$$(10) \quad \tau c \tau^{-1} = e_{01}(q) e_{34}(1).$$

The second factor commutes with  $\ell = e_{12}(1)$ . The identity  $[e_{ij}(a), e_{jk}(b)] = e_{ik}(ab)$  therefore gives  $d = e_{02}(q)$ , which is nontrivial because  $q \neq 0$ . Its normal closure is a nontrivial normal subgroup of  $H$ , and hence is all of  $H$  by Proposition 3.3.  $\square$

*Proof of Theorem B.* The ring  $R$  is a finitely generated  $\mathbb{F}_2$ -algebra, so  $H$  is countable, and  $H$  has property (T) by the Ershov–Jaikin–Zapirain theorem for elementary groups over finitely generated rings [EJZ, Theorem 1.1].

The containment (9), the centrality of  $c$  relative to  $L$ , and Proposition 3.4 show that  $d \in \mathfrak{D}_H(L)$ . The subgroup  $\mathfrak{D}_H(L)$  is normal and Proposition 3.4 gives  $\langle\langle d \rangle\rangle_H = H$ , so  $\mathfrak{D}_H(L) = H$ . The group  $H$  has property (T), so Theorem A, applied with  $K = H$ , gives  $\text{Rad}_{\text{MF}}(H) = H$ . Proposition 3.3 shows that  $H$  is nontrivial and simple.

If  $M$  is MF and  $f: H \rightarrow M$  is a homomorphism, compose  $f$  with a faithful corona embedding of  $M$ . Since  $\text{Rad}_{\text{MF}}(H) = H$ , this composition and hence  $f$  are trivial. If  $H$  were MF, its identity homomorphism would contradict this conclusion.

Finally,  $C_r^*(H)$  is separable because  $H$  is countable. Its canonical trace is faithful, so it is stably finite. If an MF embedding  $e: C_r^*(H) \rightarrow \mathcal{Q}_{\mathbf{d}}$  existed and  $p = e(1)$ , then  $h \mapsto e(\lambda_h) + (1 - p)$  would embed  $H$  in  $\mathcal{U}(\mathcal{Q}_{\mathbf{d}})$ . This contradicts the fact that  $H$  is not MF.  $\square$

#### 4. OPERATOR NORM MODELS ON FINITE SETS

The MF property is defined through sequences of models on all of  $G$ , but it is determined by finite data with a fixed separation constant. This section proves that reformulation and derives the upper bound of Theorem C from it.

**Lemma 4.1** (locality with fixed separation). *A countable group  $G$  is MF if and only if for every finite set  $F \subseteq G$  containing 1 and every  $\varepsilon > 0$  there are  $d \geq 1$  and a map  $V: F \rightarrow \mathcal{U}(d)$  with  $V(1) = 1$  such that*

$$\begin{aligned} \|V(gh) - V(g)V(h)\| &\leq \varepsilon \quad \text{whenever } g, h, gh \in F, \\ \|V(g) - 1\| &\geq 1 \quad \text{for } g \in F \setminus \{1\}. \end{aligned}$$

*Proof.* Suppose first that every finite  $F \ni 1$  and every  $\varepsilon > 0$  admit such a map. Enumerate  $G = \{g_0 = 1, g_1, g_2, \dots\}$ , let  $F_n = \{g_i g_j : i, j \leq n\} \cup \{g_i : i \leq n\}$ , let  $V_n: F_n \rightarrow \mathcal{U}(d_n)$  be a map for  $(F_n, 1/n)$ , and extend it to  $G$  by  $V_n(g) = 1$  for  $g \notin F_n$ . For  $g, h \in G$  the elements  $g, h, gh$  lie in  $F_n$  for

all large  $n$ , so  $\|V_n(gh) - V_n(g)V_n(h)\| \leq 1/n \rightarrow 0$ ; and for  $g \neq 1$  we have  $\|V_n(g) - 1\| \geq 1$  for all large  $n$ . Hence  $G$  is MF.

Conversely, let  $V_n: G \rightarrow \mathcal{U}(d_n)$  be MF models:  $V_n(1) = 1$ , the multiplication defects tend to zero, and  $c_g = \limsup_n \|V_n(g) - 1\| > 0$  for every  $g \neq 1$ . The separation constants  $c_g$  may be arbitrarily small, and we first amplify them. For a unitary  $u$  and  $k \geq 1$  the tensor power  $u^{\otimes k}$  is a unitary of dimension  $d^k$ ,  $(uv)^{\otimes k} = u^{\otimes k}v^{\otimes k}$ , and telescoping gives  $\|u^{\otimes k} - v^{\otimes k}\| \leq k\|u - v\|$ . If  $\|u - 1\| = c > 0$ , write  $c = 2\sin(\theta/2)$  with  $0 < \theta \leq \pi$ ; then  $e^{i\theta}$  or  $e^{-i\theta}$  is an eigenvalue of  $u$ , so  $e^{ik\theta}$  or  $e^{-ik\theta}$  is an eigenvalue of  $u^{\otimes k}$ , and  $\|u^{\otimes k} - 1\| \geq |e^{ik\theta} - 1| = 2|\sin(k\theta/2)|$ . If  $\theta \geq \pi/2$  take  $k = 1$ ; otherwise the first  $k$  with  $k\theta \geq \pi/2$  satisfies  $k\theta < \pi/2 + \theta < \pi$ . In both cases  $k\theta \in [\pi/2, 3\pi/2]$  modulo  $2\pi$ , so  $\|u^{\otimes k} - 1\| \geq \sqrt{2}$ . Write  $k(c)$  for this exponent; it depends only on  $c$ .

Now fix a finite  $F \ni 1$  and  $\varepsilon > 0$ , let  $K = \max_{g \in F \setminus \{1\}} k(c_g)$ , and put

$$W_n(g) = \bigoplus_{k=1}^K V_n(g)^{\otimes k} \in \mathcal{U}(D_n), \quad D_n = \sum_{k=1}^K d_n^k.$$

Then  $W_n(1) = 1$  and  $\|W_n(gh) - W_n(g)W_n(h)\| \leq K\|V_n(gh) - V_n(g)V_n(h)\|$ , so the defects of  $W_n$  tend to zero. For  $g \in F \setminus \{1\}$  choose a subsequence along which  $\|V_n(g) - 1\| \rightarrow c_g$  and eigenvalues  $\lambda_n$  of  $V_n(g)$  with  $|\lambda_n - 1| \rightarrow c_g$ ; passing to a further subsequence,  $\lambda_n \rightarrow \lambda$  with  $|\lambda - 1| = c_g$ , and then  $\lambda_n^k \rightarrow \lambda^k$  with  $|\lambda^k - 1| \geq \sqrt{2}$  for  $k = k(c_g)$ . Since  $\lambda_n^k$  is an eigenvalue of  $V_n(g)^{\otimes k}$ , it follows that  $\limsup_n \|W_n(g) - 1\| \geq \sqrt{2} > 1$ . Choose  $n_0$  such that for  $n \geq n_0$  every defect of  $W_n$  on  $F$  is at most  $\varepsilon$ , and for each  $g \in F \setminus \{1\}$  choose  $m_g \geq n_0$  with  $\|W_{m_g}(g) - 1\| \geq 1$ . Let  $V$  be the restriction to  $F$  of  $\bigoplus_{g \in F \setminus \{1\}} W_{m_g}$  (and of  $W_{n_0}$  if  $F = \{1\}$ ). Then  $V(1) = 1$ , the defect of  $V$  on  $F$  is the maximum of the defects of its blocks, hence at most  $\varepsilon$ , and for  $g \in F \setminus \{1\}$  the block  $m_g$  gives  $\|V(g) - 1\| \geq 1$ .  $\square$

**Lemma 4.2** (marked closedness). *The class of MF groups is closed in the space of marked groups. In particular, every finitely generated group that is not MF is a quotient of a finitely presented group that is not MF.*

*Proof.* An  $(F, \varepsilon)$ -model in the sense of Lemma 4.1 depends only on the multiplication table of  $F$ , that is, on which products  $gh$  with  $g, h \in F$  lie in  $F$  and which elements of  $F$  are equal. If  $G = \langle X \mid N \rangle$  is finitely generated and not MF, Lemma 4.1 gives a finite  $F \ni 1$  and  $\varepsilon > 0$  with no  $(F, \varepsilon)$ -model. Represent the elements of  $F$  by words in  $X$ ; the finitely many equalities among the words and their pairwise products that hold in  $G$  are consequences of finitely many relators  $N_0 \subseteq N$ , and the inequalities hold in every marked group mapping onto  $G$ . Hence every marked group that agrees with  $G$  on the finitely many words involved carries a table isomorphic to that of  $F$ , and so is not MF; this is closedness. Applied to  $\langle X \mid N_0 \rangle$ , which maps onto  $G$  and carries the same table, it gives the finitely presented quotient.  $\square$

We use the Kleene–Mostowski arithmetical hierarchy in its standard form [So]: a set  $A \subseteq \mathbb{N}$  is  $\Sigma_1^0$  if  $A = \{x : \exists y R(x, y)\}$  for a decidable relation  $R$ , it is  $\Pi_1^0$  if its complement is  $\Sigma_1^0$ , and  $\Sigma_2^0$  and  $\Pi_2^0$  are the classes  $\{x : \exists y \forall z R(x, y, z)\}$  and  $\{x : \forall y \exists z R(x, y, z)\}$  with  $R$  decidable. Fix an effective coding of finite presentations by natural numbers, under which a code determines computably its number of generators and its finite list of relator words, and write  $G_P$  for the group presented by the code  $P$ . Let  $\text{MF}_{\text{fp}}$  be the set of codes  $P$  such that  $G_P$  is MF, and  $\text{NONMF}_{\text{fp}}$  its complement.

**Proposition 4.3** (the certificate normal form). *There is a decidable relation  $C(P, n, c)$  on triples of natural numbers such that a code  $P$  lies in  $\text{MF}_{\text{fp}}$  if and only if for every  $n$  there is  $c$  with  $C(P, n, c)$ . Consequently  $\text{MF}_{\text{fp}} \in \Pi_2^0$  and  $\text{NONMF}_{\text{fp}} \in \Sigma_2^0$ .*

*Proof.* Let  $P = \langle x_1, \dots, x_k \mid r_1, \dots, r_m \rangle$ . A *certificate at scale  $n$*  is a triple  $c = (d, \ell, \pi)$  consisting of a dimension  $d \geq 1$ , a labelling  $\ell$  of every reduced word of length at most  $n$  in  $x_1^{\pm 1}, \dots, x_k^{\pm 1}$  by one of the symbols **T** and **S**, and, for every word  $w$  labelled **T**, an expression of  $w$  as a product of conjugates of the relators and their inverses that freely reduces to  $w$ . Such an expression is checkable, and its existence forces  $w = 1$  in  $G_P$ . Let  $\Phi(P, n, c)$  be the statement that there are  $U_1, \dots, U_k \in \mathcal{U}(d)$  with  $\|r_i(U) - 1\| \leq 2^{-n}$  for  $i \leq m$  and  $\|w(U) - 1\| \geq 1/4$  for every word  $w$  labelled **S**, where  $w(U)$  is the evaluation of  $w$  at  $U_1, \dots, U_k$  with  $x_j^{-1} \mapsto U_j^*$ . Writing  $U_j = A_j + iB_j$  with real matrices, unitarity is a system of polynomial equations, and a bound  $\|X\| \leq t$  or  $\|X\| \geq t$  on a matrix whose entries are polynomials in the variables is a semialgebraic condition, since  $\|X\| \leq t$  holds precisely when  $t^2 - X^*X$  is positive semidefinite. Thus  $\Phi(P, n, c)$  is a sentence of the first-order theory of the ordered field of real numbers, and its truth is decidable by Tarski’s theorem [Ta]. Let  $C(P, n, c)$  hold when  $c$  is a well-formed certificate at scale  $n$  whose **T**-expressions check and  $\Phi(P, n, c)$  holds. Then  $C$  is decidable.

Suppose that  $G_P$  is MF. Fix  $n$ , let  $L = \max(n, |r_1|, \dots, |r_m|)$ , let  $F$  be the set of images in  $G_P$  of all words of length at most  $L$ , together with their inverses, and let  $\varepsilon = 2^{-n}/(4L)$ . Lemma 4.1 gives  $V: F \rightarrow \mathcal{U}(d)$  with  $V(1) = 1$ , defect at most  $\varepsilon$  on  $F$ , and  $\|V(g) - 1\| \geq 1$  for  $g \in F \setminus \{1\}$ . Put  $U_j = V(\bar{x}_j)$ , where  $\bar{w}$  denotes the image of a word  $w$  in  $G_P$ . For  $g \in F$  we have  $g^{-1} \in F$  and  $\|V(g)V(g^{-1}) - 1\| \leq \varepsilon$ , so  $\|V(g^{-1}) - V(g)^*\| \leq \varepsilon$ . For a word  $w = y_1 \cdots y_\ell$  of length  $\ell \leq L$ , all of whose prefixes have images in  $F$ , induction on  $\ell$  gives  $\|w(U) - V(\bar{w})\| \leq 2\ell\varepsilon$ . In particular  $\|r_i(U) - 1\| \leq 2L\varepsilon \leq 2^{-n}$ . Label a word  $w$  of length at most  $n$  by **T** if  $\bar{w} = 1$ , with a normal-closure expression as witness, and by **S** otherwise; for the latter,  $\|w(U) - 1\| \geq \|V(\bar{w}) - 1\| - 2n\varepsilon \geq 1 - \frac{1}{2} \geq \frac{1}{4}$ . So  $C(P, n, c)$  holds for this certificate.

Conversely, suppose that for every  $n$  some certificate  $c_n = (d_n, \ell_n, \pi_n)$  satisfies  $C(P, n, c_n)$ , and choose  $U^{(n)} \in \mathcal{U}(d_n)^k$  witnessing  $\Phi(P, n, c_n)$ . For each  $g \in G_P$  fix a reduced word  $w_g$  representing  $g$ , with  $w_1$  the empty



word, and put  $V_n(g) = w_g(U^{(n)})$ . Then  $V_n(1) = 1$ . For  $g, h \in G_P$  the word  $t = w_g w_h w_{gh}^{-1}$  is trivial in  $G_P$ , so it is freely equal to a product of  $A$  conjugates of relators and their inverses, for some  $A$  depending on  $g$  and  $h$  but not on  $n$ . Since evaluation is a homomorphism on the free group, since the operator norm is unitarily invariant, and since  $\|xy - 1\| \leq \|x - 1\| + \|y - 1\|$  for unitaries  $x, y$ ,

$$\|V_n(g)V_n(h) - V_n(gh)\| = \|t(U^{(n)}) - 1\| \leq A 2^{-n} \longrightarrow 0.$$

For  $g \neq 1$  the word  $w_g$  is nontrivial in  $G_P$ , so for  $n \geq |w_g|$  the certificate  $c_n$  cannot label it **T**, hence labels it **S**, and  $\|V_n(g) - 1\| \geq 1/4$ . Thus  $\limsup_n \|V_n(g) - 1\| \geq 1/4 > 0$ , and  $G_P$  is MF.

The displayed equivalence exhibits  $\text{MF}_{\text{fp}}$  as  $\{P : \forall n \exists c C(P, n, c)\}$ , a  $\Pi_2^0$  set, and its complement as a  $\Sigma_2^0$  set.  $\square$

## 5. HNN EXTENSIONS WITH A CORONA CONJUGATOR

The preceding sections turned a one-sided conjugation inside a norm matrix corona into an obstruction to operator norm approximation. Theorem C also needs the complementary positive principle: when the conjugation is implemented by a unitary of the corona, an HNN extension remains MF. The central case  $\theta = \text{id}$  is close to Shulman's permanence theorem for central HNN extensions [Sh]; the form needed below carries a tracial state that vanishes off the identity, which is what makes the group inject, and allows an edge isomorphism implemented by a corona unitary. Its hypothesis is accordingly stronger than the MF property of the base.

A countable group  $G$  admits a *tracial MF realization*  $(A, \rho, \tau)$  if  $A$  is a separable unital MF  $C^*$ -algebra,  $\rho: G \rightarrow \mathcal{U}(A)$  is a homomorphism, and  $\tau$  is a tracial state on  $A$  with  $\tau(\rho(g)) = 0$  for every  $g \neq 1$ . Since  $\tau(1) = 1$ , such a group embeds in  $\mathcal{U}(A)$ , so it is MF. Padding blocks with zeros embeds every norm matrix corona  $\mathcal{Q}_{\mathbf{d}}$  isometrically into  $\prod_n M_n(\mathbb{C}) / \bigoplus_n M_n(\mathbb{C})$ , so the MF algebras of the introduction are exactly the separable  $C^*$ -algebras that embed in the latter corona, as in [BK, Sh].

**Lemma 5.1** (residually finite groups). *Every countable residually finite group admits a tracial MF realization.*

*Proof.* Let  $G$  be countable and residually finite, and choose finite-index normal subgroups  $N_1 \geq N_2 \geq \dots$  with  $\bigcap_k N_k = \{1\}$ ; this is possible because  $G$  is countable. Let  $d_k = [G : N_k]$  and let  $\lambda_k: G \rightarrow \mathcal{U}(d_k)$  be the left regular representation of  $G/N_k$ . In the corona  $\mathcal{Q}_{\mathbf{d}}$  put  $\rho(g) = [(\lambda_k(g))_k]$ ; then  $\rho$  is a homomorphism into the unitary group. Fix a free ultrafilter  $\omega$  on  $\mathbb{N}$  and set  $\tau([(x_k)_k]) = \lim_{\omega} \text{tr}_{d_k}(x_k)$ , where  $\text{tr}_{d_k}$  is the normalized trace on  $M_{d_k}(\mathbb{C})$ ; this is well defined because  $|\text{tr}_{d_k}(x_k)| \leq \|x_k\|$ , and it is a tracial state on  $\mathcal{Q}_{\mathbf{d}}$ . If  $g \neq 1$ , then  $g \notin N_k$  for all large  $k$ , so  $\text{tr}_{d_k}(\lambda_k(g)) = 0$  for all large  $k$  and  $\tau(\rho(g)) = 0$ . Let  $A$  be the  $C^*$ -algebra generated by  $\rho(G)$  and the unit of  $\mathcal{Q}_{\mathbf{d}}$ ; it is separable, unital, and MF, and  $(A, \rho, \tau|_A)$  is a tracial MF realization of  $G$ .  $\square$

Let  $G$  be a countable group, let  $S \leq G$  be a subgroup, and let  $\theta: S \rightarrow G$  be an injective homomorphism. The HNN extension

$$(11) \quad R = \langle G, t \mid tst^{-1} = \theta(s) \ (s \in S) \rangle$$

contains  $G$  by Britton's lemma [Bri].

**Theorem 5.2** (HNN permanence with a corona conjugator). *In the situation of (11), suppose that  $G$  admits a tracial MF realization  $(A, \rho, \tau)$ , and that for some norm matrix corona  $\mathcal{Q}$  there are an injective  $*$ -homomorphism  $\iota: A \rightarrow \mathcal{Q}$  and a unitary  $W \in \mathcal{Q}$  with*

$$W \iota \rho(s) W^* = \iota \rho(\theta(s)) \quad (s \in S).$$

*Then  $R$  admits a tracial MF realization. In particular,  $R$  is MF.*

*Proof.* Write  $D = C^*(\iota \rho(G)) \subseteq \mathcal{Q}$ , a separable  $C^*$ -algebra with unit  $p = \iota(1)$ , and let  $B_0 = C^*(\iota \rho(S))$  and  $B_1 = C^*(\iota \rho(\theta(S)))$ , both containing  $p$ . Conjugation by  $W$  restricts to a  $*$ -isomorphism  $\Theta: B_0 \rightarrow B_1$  carrying  $\iota \rho(s)$  to  $\iota \rho(\theta(s))$ . Let  $U$  be the quotient of the full unital free product  $D * C(\mathbb{T})$  by the closed ideal generated by the elements  $ubu^* - \Theta(b)$ ,  $b \in B_0$ , where  $u$  is the canonical unitary generator of  $C(\mathbb{T})$ . We write  $d$  and  $u$  for the images of  $d \in D$  and  $u$  in  $U$ . By construction,  $U$  has the following universal property: whenever  $\pi: D \rightarrow E$  is a unital  $*$ -homomorphism into a unital  $C^*$ -algebra and  $v \in \mathcal{U}(E)$  satisfies  $v\pi(b)v^* = \pi(\Theta(b))$  for  $b \in B_0$ , there is a unique  $*$ -homomorphism  $U \rightarrow E$  extending  $\pi$  and sending  $u$  to  $v$ . We first embed  $U$  in an amalgamated free product, which Shulman's criterion shows to be MF, and then construct the required tracial state on the copy of  $R$  inside  $U$ .

*Step 1:  $U$  embeds in an amalgamated free product.* This step is a self-contained form of a corner argument of Ueda [Ue]. Let  $C = B_0 \oplus B_1$ ,  $A_1 = M_2(D)$ , and  $A_2 = M_2(B_0)$ , with the unital inclusions

$$\iota_A(c_0, c_1) = \begin{pmatrix} c_0 & 0 \\ 0 & c_1 \end{pmatrix} \in A_1, \quad \iota_B(c_0, c_1) = \begin{pmatrix} c_0 & 0 \\ 0 & \Theta^{-1}(c_1) \end{pmatrix} \in A_2,$$

and let  $P = A_1 *_C A_2$  be the full amalgamated free product, in which  $A_1$  and  $A_2$  embed [Sh, Section 2.1]. Write  $e_{ij}$  and  $f_{ij}$  for the matrix units of  $A_1$  and  $A_2$ . Since  $\iota_A(p, 0) = \iota_B(p, 0)$  and  $\iota_A(0, p) = \iota_B(0, p)$ , we have  $e_{11} = f_{11} =: e$  and  $e_{22} = f_{22}$  in  $P$ . Define  $\pi_D: D \rightarrow ePe$  by  $\pi_D(d) = \text{diag}(d, 0)$ , a unital  $*$ -homomorphism into the corner, and put  $w = e_{12}f_{21} \in ePe$ . Then  $ww^* = e_{12}f_{22}e_{21} = e_{12}e_{22}e_{21} = e$  and  $w^*w = f_{12}e_{22}f_{21} = f_{12}f_{22}f_{21} = e$ , so  $w$  is a unitary of  $ePe$ . For  $b \in B_0$  the element  $\text{diag}(b, 0)$  is  $\iota_A(b, 0) = \iota_B(b, 0)$ , so

$$\begin{aligned} w\pi_D(b)w^* &= e_{12}(f_{21} \text{diag}(b, 0)f_{12})e_{21} = e_{12} \iota_B(0, \Theta(b)) e_{21} \\ &= e_{12} \iota_A(0, \Theta(b)) e_{21} = \pi_D(\Theta(b)), \end{aligned}$$

where the middle equality uses  $\iota_B(0, \Theta(b)) = \text{diag}(0, b)$  in  $A_2$ . The universal property of  $U$  gives a  $*$ -homomorphism  $\Phi: U \rightarrow ePe \subseteq P$  with  $\Phi|_D = \pi_D$  and  $\Phi(u) = w$ . We prove that  $\Phi$  is injective by constructing  $\Psi: P \rightarrow M_2(U)$  with  $\Psi\Phi(x) = \text{diag}(x, 0)$ . Let  $\Psi_1: A_1 \rightarrow M_2(U)$  be the entrywise inclusion  $D \subseteq U$ , and let  $\Psi_2: A_2 \rightarrow M_2(U)$  be the entrywise inclusion  $B_0 \subseteq U$  followed

by conjugation by  $\text{diag}(1, u)$ . On  $C$  we have  $\Psi_1(\iota_A(c_0, c_1)) = \text{diag}(c_0, c_1)$  and  $\Psi_2(\iota_B(c_0, c_1)) = \text{diag}(c_0, u\Theta^{-1}(c_1)u^*) = \text{diag}(c_0, c_1)$ , so the pair induces a  $*$ -homomorphism  $\Psi: P \rightarrow M_2(U)$ . On generators,  $\Psi\Phi(d) = \text{diag}(d, 0)$  and  $\Psi\Phi(u) = \Psi_1(e_{12})\Psi_2(f_{21}) = \text{diag}(u, 0)$ . Hence  $\Psi\Phi$  is the injective  $*$ -homomorphism  $x \mapsto \text{diag}(x, 0)$  of  $U$  into  $M_2(U)$ , and  $\Phi$  is injective.

*Step 2:  $P$  is MF.* By Shulman's criterion [Sh, Theorem 16], the full amalgamated free product  $A_1 *_C A_2$  of separable  $C^*$ -algebras is MF as soon as there are injective  $*$ -homomorphisms  $\varphi_A$  and  $\varphi_B$  of  $A_1$  and  $A_2$  into one norm matrix corona with  $\varphi_A \circ \iota_A = \varphi_B \circ \iota_B$ . Identify  $M_2(\mathcal{Q})$  with the norm matrix corona of the doubled dimension sequence, and let  $\varphi_A: M_2(D) \rightarrow M_2(\mathcal{Q})$  be the entrywise inclusion and  $\varphi_B: M_2(B_0) \rightarrow M_2(\mathcal{Q})$  the entrywise inclusion followed by conjugation by  $\text{diag}(1, W)$ . Both are injective, and on  $C$ ,

$$\varphi_B(\iota_B(c_0, c_1)) = \begin{pmatrix} c_0 & 0 \\ 0 & W\Theta^{-1}(c_1)W^* \end{pmatrix} = \begin{pmatrix} c_0 & 0 \\ 0 & c_1 \end{pmatrix} = \varphi_A(\iota_A(c_0, c_1)),$$

because  $\Theta$  is conjugation by  $W$ . Hence  $P$  is MF, and so are its  $C^*$ -subalgebras  $ePe$  and  $\Phi(U) \cong U$ .

*Step 3: the tracial state.* The functional  $T = \tau \circ \iota^{-1}$  is a tracial state on  $\iota(A) \supseteq D$  with  $T(\iota\rho(g)) = 0$  for  $g \neq 1$ . In the GNS representation of  $D$  for  $T$ , the vectors  $\xi_g = \iota\rho(g)\xi_T$ ,  $g \in G$ , are orthonormal, because  $\langle \xi_g, \xi_h \rangle = T(\iota\rho(h^{-1}g))$ . Their closed span is invariant under  $D$ , and on it  $\iota\rho(g)$  acts as the left regular representation  $\lambda_G(g)$  of  $G$ . Thus there is a  $*$ -homomorphism  $\pi: D \rightarrow C_r^*(G)$  with  $\pi(\iota\rho(g)) = \lambda_G(g)$ . Since  $G \leq R$ , the space  $\ell^2(R)$  is a direct sum of copies of  $\ell^2(G)$  indexed by the right cosets of  $G$ , so  $\lambda_R|_G$  is a multiple of  $\lambda_G$  and the assignment  $\lambda_G(g) \mapsto \lambda_R(g)$  extends to a  $*$ -homomorphism  $C_r^*(G) \rightarrow C_r^*(R)$ . Composing, we obtain  $\sigma_0: D \rightarrow C_r^*(R)$  with  $\sigma_0(\iota\rho(g)) = \lambda_R(g)$ . The pair  $(\sigma_0, \lambda_R(t))$  satisfies the covariance relation  $\lambda_R(t)\sigma_0(b)\lambda_R(t)^* = \sigma_0(\Theta(b))$  for  $b \in B_0$ : on the generators  $\iota\rho(s)$  both sides equal  $\lambda_R(\theta(s))$ , because  $tst^{-1} = \theta(s)$  in  $R$ , and two  $*$ -homomorphisms that agree on generators agree on  $B_0$ . The universal property of  $U$  gives  $\sigma: U \rightarrow C_r^*(R)$  with  $\sigma(\iota\rho(g)) = \lambda_R(g)$  and  $\sigma(u) = \lambda_R(t)$ .

Define  $j: R \rightarrow \mathcal{U}(U)$  by  $j(g) = \iota\rho(g)$  for  $g \in G$  and  $j(t) = u$ . The defining relations of (11) hold in  $U$ , since  $u\iota\rho(s)u^* = \Theta(\iota\rho(s)) = \iota\rho(\theta(s))$ , so  $j$  is a homomorphism, and  $\sigma \circ j = \lambda_R$  is injective, so  $j$  is injective. Let  $A' = C^*(j(R)) \subseteq U$ , a separable unital MF  $C^*$ -algebra, and let  $\tau' = \tau_R \circ \sigma|_{A'}$ , where  $\tau_R$  is the canonical trace  $x \mapsto \langle x\delta_1, \delta_1 \rangle$  of  $C_r^*(R)$ . Then  $\tau'$  is a tracial state on  $A'$  with  $\tau'(j(r)) = \tau_R(\lambda_R(r)) = 0$  for  $r \neq 1$ . Hence  $(A', j, \tau')$  is a tracial MF realization of  $R$ .  $\square$

The covariant pair  $(\iota\rho, W)$  need not define a faithful representation of  $R$  in  $\mathcal{Q}$ , and in the applications below it does not; the map  $\sigma: U \rightarrow C_r^*(R)$  is what proves that  $j: R \rightarrow \mathcal{U}(U)$  is injective. The proof uses no normal form for the HNN extension.

**Corollary 5.3** (central HNN extensions). *If  $G$  admits a tracial MF realization and  $S \leq G$  is any subgroup, then  $\langle G, t \mid [t, s] = 1 \ (s \in S) \rangle$  admits a tracial MF realization.*

*Proof.* Apply Theorem 5.2 with  $\theta$  the inclusion of  $S$ , with  $\iota$  any injective  $*$ -homomorphism of  $A$  into a norm matrix corona, and with  $W = 1$ .  $\square$

## 6. THE COMPLEXITY OF RECOGNIZING MF GROUPS

Write  $W_e$  for the domain of the  $e$ -th partial computable function and

$$\text{INF} = \{e : W_e \text{ is infinite}\}, \quad \text{FIN} = \{e : W_e \text{ is finite}\}.$$

The set INF is  $\Pi_2^0$ -complete and FIN is  $\Sigma_2^0$ -complete [So, Chapter IV]. We now combine the obstruction of Theorem B with the permanence of Theorem 5.2: this section constructs a computable map  $e \mapsto \hat{R}_e$  into finite presentations such that the presented group contains  $H$  when  $e \in \text{FIN}$  and falls under Theorem 5.2 when  $e \in \text{INF}$ , so that it is MF if and only if  $e \in \text{INF}$ . The construction of Bilanovic–Chubb–Roven gives a recursively presented group  $C_e$  that is trivial when  $e \in \text{INF}$  and contains  $H$  when  $e \in \text{FIN}$ . We embed  $C_e$  in a three-generated recursively presented group  $Q_e = F/N_e$ , use a Mikhailova subgroup to detect  $N_e$  inside a finitely presented group, and apply two HNN extensions, in the manner of Higman’s rope trick [Hig], to obtain  $\hat{R}_e$ . This section proves that  $\hat{R}_e$  is not MF when  $e \in \text{FIN}$ ; Section 7 proves that it is MF when  $e \in \text{INF}$ .

### A recursive presentation of $H$ .

**Lemma 6.1** (a recursive presentation of  $H$ ). *The group  $H = \text{EL}_{12}(R)$  is finitely generated and has decidable word problem. In particular, it has a recursive presentation on finitely many generators.*

*Proof.* Let  $Y$  be the set of elements  $e_{ij}(a)$  with  $i \neq j$  and  $a \in \{1, s_0, s_1, t_0, t_1\}$ . Since  $e_{ij}(a + b) = e_{ij}(a)e_{ij}(b)$  and  $[e_{ik}(a), e_{kj}(b)] = e_{ij}(ab)$  for distinct  $i, j, k$ , and since every element of  $R$  is a sum of monomials in the generators  $s_0, s_1, t_0, t_1$ , induction on the length of a monomial shows that every  $e_{ij}(a)$  lies in the subgroup generated by  $Y$ ; there is always an index  $k \notin \{i, j\}$  because  $12 \geq 3$ . Thus  $H = \langle Y \rangle$ . The monomials in  $s_0, s_1, t_0, t_1$  that contain no subword  $t_i s_j$  and no subword  $s_1 t_1$  form the standard reduced  $\mathbb{F}_2$ -basis of  $R$  [Lev]. Products of reduced monomials are computed by the defining reductions, so equality in  $R$  is decidable. A word in  $Y^{\pm 1}$  is therefore evaluated to an explicit  $12 \times 12$  matrix over  $R$ , and it represents 1 in  $H$  exactly when that matrix is the identity. The set of words representing 1 is decidable, hence recursively enumerable, and  $\langle Y \mid \text{that set} \rangle$  is a recursive presentation of  $H$ .  $\square$

**The group  $C_e$ .**

**Lemma 6.2** (the group  $C_e$ ). *There is an algorithm that computes from  $e$  a recursive presentation, on a computable set of generators, of a group  $C_e$  such that  $C_e$  is trivial if  $e \in \text{INF}$  and  $H$  embeds in  $C_e$  if  $e \in \text{FIN}$ .*

*Proof.* This is the construction of Bilanovic–Chubb–Roven [BCR, Theorem 3.1] with the trivial group as positive witness and  $H$  as negative witness. Let  $\langle Y \mid T \rangle$  be the recursive presentation of  $H$  from Lemma 6.1. Take generators  $y_{i,\ell}$  for  $i \geq 1$  and  $\ell \in Y$ , one copy  $Y_i$  of  $Y$  for each  $i$ , and as relators all words of  $T$  rewritten in each copy  $Y_i$ , together with every generator  $y_{i,\ell}$  for which  $i \leq |W_e|$ . The last family is recursively enumerable uniformly in  $e$ : enumerate  $W_e$ , and whenever its  $i$ -th element appears, enumerate all of  $Y_i$ . The group presented is the free product of the copies  $H_i \cong H$ ,  $i \geq 1$ , with the first  $|W_e|$  copies killed. If  $W_e$  is infinite every copy is killed and  $C_e$  is trivial; if  $W_e$  is finite,  $C_e$  is the free product of the remaining copies and contains  $H$ .  $\square$

**A three-generator embedding.** Let  $F = F(x, y, t)$  be the free group on  $x, y, t$ , and let  $y_i = x^i y x^{-i}$  for  $i \in \mathbb{Z}$ . For a countable group  $C$  with a generating sequence  $(c_i)_{i \geq 1}$ , put  $c_i = 1$  for  $i \leq 0$  and

$$(12) \quad B(C) = \langle C * F(x, y), t \mid ty_i t^{-1} = c_i y_i \ (i \in \mathbb{Z}) \rangle.$$

Let  $H_0 = B(1) = \langle x, y, t \mid [t, y_i] = 1 \ (i \in \mathbb{Z}) \rangle$ , let  $q_+ : F \rightarrow H_0$  be the quotient map, and let  $N_+ = \ker q_+$ .

**Lemma 6.3** (the three-generator embedding).

- (1)  $B(C)$  is an HNN extension of  $C * F(x, y)$ , so  $C$  embeds in  $B(C)$ ; it is generated by  $x, y, t$ ; and a recursive presentation of  $B(C)$  on  $x, y, t$  is computable from a recursive presentation of  $C$ . We write  $B(C) = F/N_C$  and  $q_C : F \rightarrow B(C)$  for the quotient map.
- (2) Killing  $C$  induces a surjection  $B(C) \rightarrow H_0$ ; hence  $N_C \leq N_+$ , with equality when  $C$  is trivial.
- (3) The assignment  $x \mapsto (x_1, x_2)$ ,  $y \mapsto (y, 1)$ ,  $t \mapsto (1, t)$  defines an injective homomorphism  $j : H_0 \rightarrow P = F(x_1, y) \times F(x_2, t)$ . In particular,  $H_0$  is residually finite.

*Proof.* (1) The elements  $y_i$ ,  $i \in \mathbb{Z}$ , freely generate the normal closure of  $y$  in  $F(x, y)$  [LS, Chapter I]. The retraction  $C * F(x, y) \rightarrow F(x, y)$  killing  $C$  sends  $c_i y_i$  to  $y_i$ , so a nontrivial reduced word in the  $c_i y_i$  maps to a nontrivial word in the  $y_i$ ; hence the  $c_i y_i$  freely generate a free subgroup, and  $y_i \mapsto c_i y_i$  is an isomorphism between two free subgroups of the base. Thus (12) is an HNN extension, and the base, hence  $C$ , embeds by Britton’s lemma. The relation  $c_i = ty_i t^{-1} y_i^{-1}$  shows that  $x, y, t$  generate  $B(C)$ , and substituting this expression for  $c_i$  into the relators of  $C$  and into (12) produces a recursive presentation on  $x, y, t$ , uniformly in the presentation of  $C$ .

(2) The relators of (12) map to the relators  $[t, y_i]$  of  $H_0$  when every  $c_i$  is sent to 1, and the map is the identity on  $x, y, t$ .

(3) The relators  $[t, y_i]$  are sent to  $[(1, t), (x_1^i y x_1^{-i}, 1)] = 1$ , so  $j$  is a homomorphism. Let  $N_0$  be the normal closure of  $\{y, t\}$  in  $H_0$ . Then  $H_0/N_0 = \langle x \rangle$  is infinite cyclic and the extension splits, so  $H_0 = N_0 \rtimes \langle x \rangle$ . Rewriting the presentation of  $H_0$  over the transversal  $\{x^n\}$  by the Reidemeister–Schreier method [LS, Chapter II] presents  $N_0$  on the generators  $y_n = x^n y x^{-n}$  and  $t_m = x^m t x^{-m}$ ,  $n, m \in \mathbb{Z}$ , with relators the conjugates  $x^m [t, y_n] x^{-m} = [t_m, y_{n+m}]$ , that is, all commutators  $[t_m, y_n]$ . Hence  $N_0 = F(y_n : n \in \mathbb{Z}) \times F(t_m : m \in \mathbb{Z})$ . Now  $j(y_n) = (x_1^n y x_1^{-n}, 1)$  and  $j(t_m) = (1, x_2^m t x_2^{-m})$ , and the elements  $x_1^n y x_1^{-n}$  freely generate the normal closure of  $y$  in  $F(x_1, y)$ , and likewise for  $t$ ; so  $j$  restricted to  $N_0$  is an isomorphism onto the product of these two normal closures. Finally  $j(x) = (x_1, x_2)$  has infinite order and  $j(x)^k \notin j(N_0)$  for  $k \neq 0$ , because the exponent sum of  $x_1$  in  $j(x^k n)$  is  $k$  for  $n \in N_0$ . Hence  $j$  is injective on  $N_0 \rtimes \langle x \rangle$ . Free groups are residually finite and residual finiteness passes to direct products and subgroups, so  $H_0$  is residually finite.  $\square$

For  $e \in \mathbb{N}$  let  $Q_e = B(C_e) = F/N_e$  with  $C_e$  the group of Lemma 6.2, and write  $q_e : F \rightarrow Q_e$ . By Lemmas 6.2 and 6.3, a recursive presentation of  $Q_e$  on  $x, y, t$  is computable from  $e$ ;  $N_e \leq N_+$ ; if  $e \in \text{INF}$  then  $Q_e = H_0$ ,  $N_e = N_+$  and  $q_e = q_+$ ; and if  $e \in \text{FIN}$  then  $H \leq C_e \leq Q_e$ .

### Detecting $N_e$ inside a finitely presented group.

**Lemma 6.4** (Higman embedding and the Mikhailova subgroup). *There is an algorithm that computes from  $e$  a finite presentation  $\langle X_e \mid R_e \rangle$  of a group  $\hat{H}_e$ , words  $w_x, w_y, w_t \in F(X_e)$ , and a finite generating set of a subgroup  $M_e \leq F(X_e) \times F(X_e)$ , such that  $x \mapsto w_x, y \mapsto w_y, t \mapsto w_t$  induces an embedding  $Q_e \rightarrow \hat{H}_e$ , and  $M_e = \{(u, v) : u = v \text{ in } \hat{H}_e\}$ . Consequently, with*

$$K_e^0 = F \times F(X_e) \times F(X_e), \quad i_e(f) = (f, w_f, 1), \quad L_e^0 = F \times M_e,$$

where  $w_f$  is the image of  $f$  under  $x \mapsto w_x, y \mapsto w_y, t \mapsto w_t$ , the map  $i_e : F \rightarrow K_e^0$  is injective and  $i_e(f) \in L_e^0$  if and only if  $f \in N_e$ .

*Proof.* Higman’s embedding theorem [Hig] embeds every finitely generated recursively presented group in a finitely presented group, and its proof is effective; an explicit algorithm producing the finite presentation and the embedding words from a recursive presentation is given by Mikaelian [Mik]. Apply it to the presentation of  $Q_e$ . Following Mikhailova [Mih], let  $M_e$  be the subgroup of  $F(X_e) \times F(X_e)$  generated by the pairs  $(z, z)$ ,  $z \in X_e$ , and  $(r, 1)$ ,  $r \in R_e$ . If  $(u, v) \in M_e$  then  $u = v$  in  $\hat{H}_e$ , since this holds for the generators. Conversely, if  $u = v$  in  $\hat{H}_e$ , then  $uv^{-1}$  lies in the normal closure of  $R_e$ , so it is a product of conjugates  $gr^{\pm 1}g^{-1}$  with  $r \in R_e$ , and  $(gr^{\pm 1}g^{-1}, 1) = (g, g)(r^{\pm 1}, 1)(g, g)^{-1} \in M_e$ ; hence  $(u, v) = (uv^{-1}, 1)(v, v) \in M_e$ . The first coordinate makes  $i_e$  injective, and  $(f, w_f, 1) \in F \times M_e$  if and only if  $w_f = 1$  in  $\hat{H}_e$ , if and only if  $q_e(f) = 1$  because the embedding is injective, if and only if  $f \in N_e$ .  $\square$

**The two HNN extensions.** Let  $P = F(x_1, y) \times F(x_2, t)$  and  $j: H_0 \rightarrow P$  be as in Lemma 6.3. Define the graph witness

$$K^g = F \times P, \quad L^g = \{(f, jq_+(f)) : f \in F\},$$

and the product witness

$$K_e = K_e^0 \times K^g, \quad L_e = L_e^0 \times L^g, \quad i: F \rightarrow K_e, \quad i(f) = (i_e(f), (f, 1)).$$

Let

$$(13) \quad \Gamma_e = \langle K_e, v \mid [v, \ell] = 1 \ (\ell \in L_e) \rangle, \quad S_e = \langle i(F), v i(F) v^{-1} \rangle \leq \Gamma_e.$$

**Lemma 6.5** (the first HNN extension).

- (1)  $K_e$  is a direct product of free groups of finite rank, hence finitely presented and residually finite;  $L_e$  is finitely generated;  $i$  is injective; and  $i(F) \cap L_e = i(N_e)$ .
- (2)  $\Gamma_e$  is finitely presented, and a finite presentation is computable from  $e$ . The natural map from the amalgamated free product  $F *_{N_e} F$ , in which  $n \in N_e$  in the first copy is identified with  $n$  in the second copy, onto  $S_e$ , sending the first copy to  $i(F)$  and the second to  $v i(F) v^{-1}$ , is an isomorphism.
- (3) There is a homomorphism  $\tau_e: S_e \rightarrow Q_e$  with  $\tau_e(i(f)) = q_e(f)$  and  $\tau_e(v i(f) v^{-1}) = 1$  for  $f \in F$ .

*Proof.* (1)  $K_e = F \times F(X_e) \times F(X_e) \times F \times P$  is a product of six free groups of finite rank, so it is finitely presented, and residually finite because free groups are. The subgroup  $L_e$  is generated by the basis of the first factor, the generators of  $M_e$  from Lemma 6.4, and the elements  $(a, jq_+(a))$  for  $a \in \{x, y, t\}$ , so it is finitely generated. Injectivity of  $i$  is clear from the first coordinate. An element  $i(f)$  lies in  $L_e$  if and only if  $i_e(f) \in L_e^0$  and  $(f, 1) \in L^g$ ; the first holds if and only if  $f \in N_e$  by Lemma 6.4, and the second if and only if  $jq_+(f) = 1$ , that is,  $f \in N_+$ . Since  $N_e \leq N_+$ , the intersection is  $i(N_e)$ .

(2) The presentation (13) is finite once  $[v, \ell] = 1$  is imposed only for the finitely many generators  $\ell$  of  $L_e$ , and it is computable from  $e$  because  $K_e$ ,  $L_e$  and  $i$  are. It is an HNN extension of  $K_e$  with both edge groups equal to  $L_e$  and the identity as edge isomorphism. The map from  $F *_{N_e} F$  to  $S_e$  is well defined because  $i(n) \in L_e$  commutes with  $v$  for  $n \in N_e$ . A reduced word of the amalgamated free product alternates between elements of the two copies of  $F$  lying outside  $N_e$ ; its image is a word  $i(f_0) v i(f_1) v^{-1} i(f_2) \cdots$  in which every letter between consecutive occurrences of  $v^{\pm 1}$  is some  $i(f_k)$  with  $f_k \notin N_e$ , hence  $i(f_k) \notin L_e$  by (1). Such a word contains no pinch  $v^{\pm 1} \ell v^{\mp 1}$  with  $\ell \in L_e$ , so it is nontrivial in  $\Gamma_e$  by Britton's lemma.

(3) Define  $\tau_e$  on  $F *_{N_e} F$  by  $q_e$  on the first copy and by the trivial map on the second; both kill  $N_e$ , so the definition is consistent on the amalgamated subgroup, and (2) transports it to  $S_e$ .  $\square$



The second HNN extension is

$$(14) \quad R_e = \langle \Gamma_e \times Q_e, u \mid u(s, 1)u^{-1} = (s, \tau_e(s)) \ (s \in S_e) \rangle,$$

whose edge maps  $s \mapsto (s, 1)$  and  $s \mapsto (s, \tau_e(s))$  are injective homomorphisms of  $S_e$  into  $\Gamma_e \times Q_e$ ; so  $\Gamma_e \times Q_e$  embeds in  $R_e$  by Britton's lemma. The presentation (14) lists the recursively enumerable relators of  $Q_e$ . Its finite replacement is

$$(15) \quad \begin{aligned} \hat{R}_e = \langle \Gamma_e \times F, u \mid & u(i(a), 1)u^{-1} = (i(a), a), \\ & u(v i(a) v^{-1}, 1)u^{-1} = (v i(a) v^{-1}, 1) \ (a \in \{x, y, t\}) \rangle, \end{aligned}$$

where  $\Gamma_e \times F$  carries the finite presentation of Lemma 6.5(2), the basis  $x, y, t$  of  $F$ , and the commutation relators between the two factors.

**Lemma 6.6** (the finite presentation). *The groups  $\hat{R}_e$  and  $R_e$  are isomorphic by the map that is the identity on  $\Gamma_e$  and on  $u$  and sends  $F$  onto  $Q_e$  by  $q_e$ . A code of the finite presentation (15) is computable from  $e$ .*

*Proof.* Both maps  $f \mapsto u(i(f), 1)u^{-1}$  and  $f \mapsto (i(f), f)$  are homomorphisms  $F \rightarrow \hat{R}_e$ , and they agree on the basis, so  $u(i(f), 1)u^{-1} = (i(f), f)$  for all  $f \in F$ ; likewise  $u(v i(f) v^{-1}, 1)u^{-1} = (v i(f) v^{-1}, 1)$  for all  $f$ . If  $n \in N_e$ , then  $i(n) \in L_e$  by Lemma 6.5(1), so  $v i(n) v^{-1} = i(n)$  in  $\Gamma_e$ , and comparing the two relations gives  $(i(n), n) = (i(n), 1)$ , that is,  $(1, n) = 1$  in  $\hat{R}_e$ . Hence the factor  $F$  of  $\hat{R}_e$  factors through  $Q_e = F/N_e$ , and the relations of (15) become  $u(s, 1)u^{-1} = (s, \tau_e(s))$  for  $s$  in the generating set  $\{i(a), v i(a) v^{-1} : a \in \{x, y, t\}\}$  of  $S_e$ , hence for all  $s \in S_e$ , since both sides are homomorphic in  $s$ . This defines a homomorphism  $R_e \rightarrow \hat{R}_e$ ; conversely, the relations of (15) hold in  $R_e$  after  $F \rightarrow Q_e$ , because  $\tau_e(i(a)) = q_e(a)$  and  $\tau_e(v i(a) v^{-1}) = 1$ . The two homomorphisms are mutually inverse on generators. Computability of the code follows from Lemmas 6.4 and 6.5.  $\square$

**Lemma 6.7** (the case  $e \in \text{FIN}$ ). *If  $e \in \text{FIN}$ , then  $H$  embeds in  $\hat{R}_e$ , and  $\hat{R}_e$  is not MF.*

*Proof.* For  $e \in \text{FIN}$  we have  $H \leq C_e \leq Q_e \leq \Gamma_e \times Q_e \leq R_e \cong \hat{R}_e$  by Lemmas 6.2, 6.3(1), 6.6 and Britton's lemma for (14). Restricting MF models to a subgroup shows that subgroups of MF groups are MF, and  $H$  is not MF by Theorem B.  $\square$

## 7. THE CASE $e \in \text{INF}$

We now prove that  $\hat{R}_e$  is MF for  $e \in \text{INF}$  and deduce Theorem C. The tracial MF realization of  $R_e$  is constructed inside a norm reduced product of MF algebras, so we first record that separable subalgebras of such products are MF. For  $C^*$ -algebras  $B_n$ ,  $n \geq 1$ , write  $\prod_n B_n / \bigoplus_n B_n$  for the quotient of the bounded sequences by the sequences with  $\|x_n\| \rightarrow 0$ ; its norm is  $\limsup_n \|x_n\|$ .



**Lemma 7.1** (reduced products of MF algebras). *If every  $B_n$  is MF, then every separable  $C^*$ -subalgebra of  $\prod_n B_n / \bigoplus_n B_n$  is MF. Moreover,  $M_k(B)$  is MF whenever  $B$  is.*

*Proof.* The second assertion holds because  $M_k$  of a norm matrix corona is the norm matrix corona of the dimension sequence multiplied by  $k$ . For the first, let  $A$  be a separable  $C^*$ -subalgebra, and choose a sequence  $(a_i)_{i \geq 1}$  dense in  $A$ , with lifts  $(a_i^{(n)})_n$ . Choose for each  $n$  an injective  $*$ -homomorphism  $\eta_n$  of  $B_n$  into a norm matrix corona, and lifts  $(a_i^{(n,k)})_k$  of  $\eta_n(a_i^{(n)})$  with  $\sup_k \|a_i^{(n,k)}\| \leq \|a_i^{(n)}\| + 1$ . Enumerate the noncommutative  $*$ -polynomials with coefficients in  $\mathbb{Q}(i)$  as  $p_1, p_2, \dots$ , and let  $\mathcal{P}_n$  be the set of those among  $p_1, \dots, p_n$  that involve only  $a_1, \dots, a_n$ , choosing  $p_1 = 0$  so that  $\mathcal{P}_n$  is nonempty. For  $p \in \mathcal{P}_n$  the number  $\|p(a^{(n)})\|$  equals  $\limsup_k \|p(a^{(n,k)})\|$ , since  $\eta_n$  is isometric. Hence there is  $K_n$  such that  $\|p(a^{(n,k)})\| \leq \|p(a^{(n)})\| + 1/n$  for all  $k \geq K_n$  and  $p \in \mathcal{P}_n$ , and for each  $p \in \mathcal{P}_n$  there is  $k_p \geq K_n$  with  $\|p(a^{(n,k_p)})\| \geq \|p(a^{(n)})\| - 1/n$ . Put

$$b_i^{(n)} = \bigoplus_{p \in \mathcal{P}_n} a_i^{(n,k_p)},$$

a block diagonal matrix. For  $p \in \mathcal{P}_n$  the norm of  $p(b^{(n)})$  is the maximum of  $\|p(a^{(n,k_{p'})})\|$  over  $p' \in \mathcal{P}_n$ , which lies between  $\|p(a^{(n)})\| - 1/n$  and  $\|p(a^{(n)})\| + 1/n$ . Every polynomial belongs to  $\mathcal{P}_n$  for all large  $n$ , so  $\limsup_n \|p(b^{(n)})\| = \limsup_n \|p(a^{(n)})\| = \|p(a)\|$  for every  $p$ . Therefore  $a_i \mapsto [(b_i^{(n)})_n]$  is well defined and isometric on the  $*$ -algebra over  $\mathbb{Q}(i)$  generated by the  $a_i$ , which is dense in  $A$ , and it extends to an isometric  $*$ -homomorphism of  $A$  into a norm matrix corona.  $\square$

**Lemma 7.2** (tensor synchronization). *Let  $\Gamma$  be a countable group with a tracial MF realization  $(A_1, \rho_1, \tau_1)$ , let  $Q$  be a countable group with homomorphisms  $\beta_n: Q \rightarrow B_n$  to finite groups such that every  $q \neq 1$  satisfies  $\beta_n(q) \neq 1$  for all large  $n$ , let  $S \leq \Gamma$ , and let  $\tau: S \rightarrow Q$  be a homomorphism. Suppose that for every  $n$  there is a homomorphism  $\lambda_n: \Gamma \rightarrow G_n$  to a finite group with*

$$\ker(\lambda_n|_S) \leq \ker(\beta_n \circ \tau).$$

*Then there are a separable unital MF  $C^*$ -algebra  $A$ , a homomorphism  $V: \Gamma \times Q \rightarrow \mathcal{U}(A)$ , a tracial state  $T$  on  $A$  with  $T(V(g, q)) = 0$  for  $(g, q) \neq (1, 1)$ , and a unitary  $W \in A$  with  $WV(s, 1)W^* = V(s, \tau(s))$  for all  $s \in S$ . Consequently*

$$\langle \Gamma \times Q, u \mid u(s, 1)u^{-1} = (s, \tau(s)) \ (s \in S) \rangle$$

*admits a tracial MF realization, and in particular is MF.*

*Proof.* Let  $E_n$  be the image of  $(\lambda_n, \beta_n): \Gamma \times Q \rightarrow G_n \times B_n$ , a finite group of order  $k_n$ , and let  $L_n$  be its left regular representation on  $\ell^2(E_n)$ . Put

$B'_n = A_1 \otimes M_{k_n}(\mathbb{C})$ , which is MF by Lemma 7.1, let  $E = \prod_n B'_n / \bigoplus_n B'_n$ , and define

$$V(g, q) = [(\rho_1(g) \otimes L_n(\lambda_n(g), \beta_n(q)))_n], \quad T([(x_n)_n]) = \lim_{\omega} (\tau_1 \otimes \text{tr}_{k_n})(x_n),$$

for a free ultrafilter  $\omega$ . Each coordinate of  $V$  is a homomorphism into a unitary group, so  $V$  is a homomorphism, and  $T$  is a tracial state on  $E$  because  $|(\tau_1 \otimes \text{tr}_{k_n})(x_n)| \leq \|x_n\|$ . If  $g \neq 1$  then  $\tau_1(\rho_1(g)) = 0$ , so  $T(V(g, q)) = 0$ ; if  $g = 1$  and  $q \neq 1$  then  $\beta_n(q) \neq 1$  for all large  $n$ , the regular representation of a finite group has trace zero off the identity, and again  $T(V(1, q)) = 0$ .

The two homomorphisms  $s \mapsto (\lambda_n(s), 1)$  and  $s \mapsto (\lambda_n(s), \beta_n \tau(s))$  from  $S$  to  $E_n$  have the same kernel, namely  $\ker(\lambda_n|_S)$ , by the hypothesis. Their images are therefore isomorphic subgroups of  $E_n$  of the same order, isomorphic via  $(\lambda_n(s), 1) \mapsto (\lambda_n(s), \beta_n \tau(s))$ . The restriction of  $L_n$  to a subgroup of  $E_n$  is unitarily equivalent to the direct sum of as many copies of the left regular representation of that subgroup as it has right cosets, so there is a unitary  $W_n \in M_{k_n}(\mathbb{C})$  with  $W_n L_n(\lambda_n(s), 1) W_n^* = L_n(\lambda_n(s), \beta_n \tau(s))$  for all  $s \in S$ . Put  $W = [(1 \otimes W_n)_n] \in E$ ; then  $WV(s, 1)W^* = V(s, \tau(s))$  for  $s \in S$ . Let  $A$  be the  $C^*$ -algebra generated by  $V(\Gamma \times Q)$  and  $W$ ; it is separable and unital, and MF by Lemma 7.1. The triple  $(A, V, T|_A)$  is a tracial MF realization of  $\Gamma \times Q$ . For the last assertion, apply Theorem 5.2 to this realization. Let  $\iota$  be an injective  $*$ -homomorphism of  $A$  into a norm matrix corona and put  $p = \iota(1)$ . The element  $\widetilde{W} = \iota(W) + (1 - p)$  is a unitary of the full corona and implements the required covariance on  $\iota(A)$ , so it is the required conjugator.  $\square$

**Lemma 7.3** (the case  $e \in \text{INF}$ ). *If  $e \in \text{INF}$ , then  $\widehat{R}_e$  admits a tracial MF realization; in particular, it is MF.*

*Proof.* Let  $e \in \text{INF}$ , so that  $Q_e = H_0$ ,  $N_e = N_+$ ,  $q_e = q_+$ , and  $\tau_e(i(f)) = q_+(f)$ ,  $\tau_e(v i(f) v^{-1}) = 1$ . By Lemma 6.6 it suffices to show that  $R_e$  admits a tracial MF realization, and by Lemma 7.2 applied to  $\Gamma = \Gamma_e$ ,  $Q = H_0$ ,  $S = S_e$  and  $\tau = \tau_e$ , it suffices to produce the data of that lemma.

The group  $K_e$  is residually finite by Lemma 6.5(1), hence admits a tracial MF realization by Lemma 5.1, and  $\Gamma_e$  is the central HNN extension of  $K_e$  over  $L_e$ , hence admits one by Corollary 5.3.

The group  $P = F(x_1, y) \times F(x_2, t)$  is residually finite; choose finite-index normal subgroups  $P_1 \geq P_2 \geq \dots$  of  $P$  with trivial intersection, let  $r_n: P \rightarrow C_n = P/P_n$ , and put  $\beta_n = r_n \circ j: H_0 \rightarrow C_n$  with  $j$  the embedding of Lemma 6.3(3). Since  $j$  is injective and the  $P_n$  decrease to the trivial group, every  $h \neq 1$  has  $\beta_n(h) \neq 1$  for all large  $n$ .

Let  $G_n = (C_n \times C_n) \rtimes \langle \sigma \rangle$ , where  $\sigma$  of order two exchanges the coordinates. Define  $\lambda_n$  on  $K_e = K_e^0 \times (F \times P)$  by killing  $K_e^0$  and sending  $(f, p)$  to  $(r_n j q_+(f), r_n(p))$ , and put  $\lambda_n(v) = \sigma$ . This respects the relations of (13): the factor  $L_e^0$  of  $L_e$  lies in  $K_e^0$  and is killed, and an element  $(f, j q_+(f))$  of  $L^g$  is sent to  $(r_n j q_+(f), r_n j q_+(f))$ , which lies on the diagonal and commutes with

$\sigma$ . Hence  $\lambda_n: \Gamma_e \rightarrow G_n$  is a homomorphism, and on the generators of  $S_e$ ,

$$\lambda_n(i(f)) = (r_n j q_+(f), 1), \quad \lambda_n(v i(f) v^{-1}) = (1, r_n j q_+(f)).$$

Let  $\pi_0, \pi_1: S_e \rightarrow H_0$  be the homomorphisms that are  $q_+$  on one copy of  $F$  in  $S_e \cong F *_{N_+} F$  and trivial on the other, as in Lemma 6.5(3); thus  $\pi_0 = \tau_e$ . The displayed formulas show that  $\lambda_n|_{S_e} = (r_n j \pi_0, r_n j \pi_1)$ , both sides being homomorphisms that agree on generators. Therefore

$$\ker(\lambda_n|_{S_e}) = \ker(r_n j \pi_0) \cap \ker(r_n j \pi_1) \leq \ker(\beta_n \circ \tau_e),$$

which is the hypothesis of Lemma 7.2.  $\square$

*Proof of Theorem C.* By Lemma 6.6 the map  $e \mapsto \hat{R}_e$  is computable, by Lemma 7.3 the presented group is MF for  $e \in \text{INF}$ , and by Lemma 6.7 it is not MF for  $e \in \text{FIN}$ . Hence  $\text{INF}$  reduces to  $\text{MF}_{\text{fp}}$  and  $\text{FIN}$  reduces to  $\text{NONMF}_{\text{fp}}$  under computable many-one reductions. Since  $\text{INF}$  is  $\Pi_2^0$ -complete and  $\text{FIN}$  is  $\Sigma_2^0$ -complete, and since  $\text{MF}_{\text{fp}} \in \Pi_2^0$  and  $\text{NONMF}_{\text{fp}} \in \Sigma_2^0$  by Proposition 4.3, the two sets are complete at their levels. A  $\Pi_2^0$ -complete set is not  $\Sigma_2^0$  and a  $\Sigma_2^0$ -complete set is not  $\Pi_2^0$ , so neither set is recursively enumerable, and in particular neither is decidable.  $\square$

The homomorphism  $V$  of Lemma 7.2 does not extend faithfully to  $R_e$  in the corona: the finite maps  $\lambda_n$  are trivial on the Mikhailova factor, so  $W$  commutes with  $V(\gamma, 1)$  for every  $\gamma \in K_e^0$ , although  $u$  does not commute with  $(\gamma, 1)$  in  $R_e$ . Injectivity is obtained in Theorem 5.2 from the universal algebra. Three questions remain open. It is not known whether the models of  $\hat{R}_e$  can be chosen to converge to  $C_r^*(\hat{R}_e)$  in operator norm, so that the reduced  $C^*$ -algebra itself is MF; the argument produces a tracial state on an abstract MF algebra, not strong convergence. It is not known whether  $\hat{R}_e$  is hyperlinear for  $e \in \text{INF}$ , which would place the recognition problem for hyperlinearity at the same level. Finally, it is not known whether an analogous family of finitely presented groups exists for soficity; the construction of Lemma 7.2 has no counterpart for permutation models.

## 8. MF RADICALS UNDER QUOTIENT MAPS

This section is not used in the proof of Theorem C. It describes what MF representations see of a group that contains a subgroup with full MF radical, and it produces every countable group as the visible quotient of a non-MF group.

**Theorem 8.1** (prescribed quotients visible to MF). *Let  $B$  be a countable group with  $\text{Rad}_{\text{MF}}(B) = B$ , and let  $1 \neq d \in B$  normally generate  $B$ . Put  $A = \langle d \rangle$  and, for a countable group  $Q$ , define*

$$W_Q = B *_A (Q \times A).$$

Then the map  $\pi_Q: W_Q \rightarrow Q$  which kills  $B$  and projects the second vertex group onto  $Q$  is a split epimorphism, and

$$\text{Rad}_{\text{MF}}(W_Q) = \pi_Q^{-1}(\text{Rad}_{\text{MF}}(Q)).$$

The group  $W_Q$  is non-MF. If  $Q$  is MF, then

$$\text{Rad}_{\text{MF}}(W_Q) = \ker \pi_Q = \langle\langle d \rangle\rangle_{W_Q}.$$

Moreover, precomposition with  $\pi_Q$  induces a bijection

$$\text{Hom}(Q, M) \longrightarrow \text{Hom}(W_Q, M)$$

for every MF group  $M$ .

We begin with the pullback of the radical along a surjection whose kernel is invisible to MF.

**Proposition 8.2** (full-kernel pullback). *Let  $f: G \rightarrow Q$  be a surjective homomorphism of countable groups. If  $\ker f \leq \text{Rad}_{\text{MF}}(G)$ , then*

$$\text{Rad}_{\text{MF}}(G) = f^{-1}(\text{Rad}_{\text{MF}}(Q)).$$

*In particular, the conclusion holds whenever  $\text{Rad}_{\text{MF}}(\ker f) = \ker f$ . More generally, for every normal subgroup  $N \trianglelefteq G$ ,*

$$\text{cl}_{\text{MF}}^G(N) = f^{-1}(\text{cl}_{\text{MF}}^Q(f(N))).$$

*Consequently,*

$$G/N \text{ is MF} \iff \ker f \leq N \text{ and } Q/f(N) \text{ is MF}.$$

*Proof.* Every corona homomorphism from  $G$  kills  $\ker f$  and therefore factors uniquely through  $f$ . Conversely, composing a corona homomorphism from  $Q$  with  $f$  gives one of  $G$ . Intersecting these two corresponding families of kernels gives the displayed MF radical pullback. For the stated sufficient condition, let  $\pi: G \rightarrow \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$  be any corona homomorphism. Its restriction to  $\ker f$  is trivial because  $\text{Rad}_{\text{MF}}(\ker f) = \ker f$ . Hence  $\ker f \leq \text{Rad}_{\text{MF}}(G)$ .

The same factorization, restricted to homomorphisms that kill  $N$ , gives the closure formula. If this closure equals  $N$ , then it contains  $\ker f$ . Mapping the equality onto  $Q$  shows that  $\text{cl}_{\text{MF}}^Q(f(N)) = f(N)$ . Conversely, these two conditions give

$$\text{cl}_{\text{MF}}^G(N) = f^{-1}(f(N)) = N.$$

The final equivalence now follows from Proposition 2.1, applied in  $G$  and  $Q$ .  $\square$

Thus image under  $f$  and inverse image under  $f$  give mutually inverse correspondences, preserving inclusion between the normal subgroups with MF quotient in  $G$  and in  $Q$ . The map induced by  $f$  on the largest quotients visible to MF is an isomorphism:

$$G/\text{Rad}_{\text{MF}}(G) \cong Q/\text{Rad}_{\text{MF}}(Q).$$

These correspondences respect composition of surjective homomorphisms with kernels invisible to MF.

Let  $B$  be a countable group with  $\text{Rad}_{\text{MF}}(B) = B$ , and let  $1 \neq d \in B$  normally generate  $B$ . Put  $A = \langle d \rangle$ . For a countable group  $Q$ , define

$$(16) \quad W_Q = B *_A (Q \times A),$$

where  $A \rightarrow Q \times A$  is the inclusion into the second factor. The trivial map  $B \rightarrow Q$  and the projection  $Q \times A \rightarrow Q$  agree on  $A$ . They therefore define an epimorphism

$$\pi_Q: W_Q \longrightarrow Q.$$

It is split by the inclusion  $Q \rightarrow Q \times A \rightarrow W_Q$ . The normal-form theorem for amalgamated free products shows that both vertex maps in (16) are injective; in particular,  $d \neq 1$  in  $W_Q$ .

**Proposition 8.3** (universal factorization). *Let  $T$  be a group such that every homomorphism  $B \rightarrow T$  is trivial. Then precomposition with  $\pi_Q$  is a bijection*

$$\text{Hom}(Q, T) \longrightarrow \text{Hom}(W_Q, T).$$

Furthermore,

$$\ker \pi_Q = \langle\langle d \rangle\rangle_{W_Q}.$$

*Proof.* Let  $f: W_Q \rightarrow T$  be a homomorphism. Its restriction to  $B$  is trivial, so its restriction to the amalgamated subgroup  $A$  is trivial. For  $(q, a) \in Q \times A$ , we then have

$$f(q, a) = f(q, 1)f(1, a) = f(q, 1).$$

Thus the restriction of  $f$  to  $Q \times A$  factors uniquely through the projection onto  $Q$ . The universal property of the amalgam shows that  $f$  factors through  $\pi_Q$ . The factorization is unique because  $\pi_Q$  is surjective.

The element  $d$  belongs to  $\ker \pi_Q$ . Conversely, quotienting  $W_Q$  by  $\langle\langle d \rangle\rangle_{W_Q}$  kills  $B$ , since  $\langle\langle d \rangle\rangle_B = B$ , and kills the  $A$ -factor of  $Q \times A$ . The resulting quotient is  $Q$ , with quotient map induced by  $\pi_Q$ . Hence  $\ker \pi_Q = \langle\langle d \rangle\rangle_{W_Q}$ .  $\square$

*Proof of Theorem 8.1.* Every homomorphism from  $B$  to an MF group is trivial: compose it with a faithful corona embedding of the target and use  $\text{Rad}_{\text{MF}}(B) = B$ . Proposition 8.3 therefore gives the asserted bijection for every MF group  $M$ . In addition, the definition of  $\text{Rad}_{\text{MF}}(B) = B$  says directly that every homomorphism from  $B$  to  $\mathcal{U}(\mathcal{Q}_{\mathbf{d}})$  is trivial. Applying Proposition 8.3 with  $T = \mathcal{U}(\mathcal{Q}_{\mathbf{d}})$  shows that every corona homomorphism from  $W_Q$  factors uniquely through  $\pi_Q$ . Intersecting their kernels gives

$$\text{Rad}_{\text{MF}}(W_Q) = \pi_Q^{-1}(\text{Rad}_{\text{MF}}(Q)).$$

The kernel of  $\pi_Q$  belongs to this radical, and it contains the nonidentity element  $d$ . Thus  $W_Q$  is non-MF. If  $Q$  is MF, its MF radical is trivial, and the displayed equality becomes

$$\text{Rad}_{\text{MF}}(W_Q) = \ker \pi_Q = \langle\langle d \rangle\rangle_{W_Q}.$$

$\square$

If  $N \trianglelefteq W_Q$ , the same factorization gives

$$(17) \quad \text{cl}_{\text{MF}}^{W_Q}(N) = \pi_Q^{-1}(\text{cl}_{\text{MF}}^Q(\pi_Q(N))).$$

Indeed, a homomorphism from  $W_Q$  to an MF group kills  $N$  exactly when its unique factor through  $Q$  kills  $\pi_Q(N)$ . Proposition 2.1 and (17) give

$$W_Q/N \text{ is MF} \iff \ker \pi_Q \leq N \text{ and } Q/\pi_Q(N) \text{ is MF.}$$

Here, if  $\ker \pi_Q \leq N$ , then  $N = \pi_Q^{-1}(\pi_Q(N))$ .

Taking  $Q = \mathbb{Z}$  makes the quotient criterion especially concrete: for every normal subgroup  $N \trianglelefteq W_{\mathbb{Z}}$ ,

$$(18) \quad W_{\mathbb{Z}}/N \text{ is MF} \iff d \in N.$$

Indeed, every quotient of  $\mathbb{Z}$  is MF, while  $\ker \pi_{\mathbb{Z}} = \langle\langle d \rangle\rangle_{W_{\mathbb{Z}}}$ .

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#### USE OF AI AND FORMAL METHODS

Claude Fable 5 and GPT-5.6 Sol assisted with mathematical exploration, proof development, literature searches, Lean formalization, drafting, and editing. The author reviewed their output and is responsible for the manuscript. Lean 4 source for the formalized results is available at <https://github.com/SauersML/group-approximation>; the repository identifies the Lean declarations corresponding to the statements below.

For this paper, the development verifies the one-sided transport theorem and the normal property-(T) radical criterion behind Theorem A; the rank 12 compression and commutator calculations and the full MF radical conclusion of Theorem B, with a direct proof of Proposition 3.3 that does not pass through Preusser’s theorem; the quotient calculus of Theorem 8.1; and, for Theorem C, the certificate normal form and the deduction of  $\Pi_2^0$ -completeness from a computable family of finite presentations that is MF exactly on INF. The construction of that family in Section 6 and Theorem 5.2 are proved in the article.

Beyond this paper, the development proves the following.

- Soficity of a recursively enumerated presentation is  $\Pi_2^0$ -complete; the hardness half follows the construction of Bilanovic–Chubb–Roven [BCR] from a finitely presented nonsoc group, whose nonsocicity is that of the Leavitt unit group of OpenAI’s chapter [OAI], re-proved in the development from Kun’s expander decomposition and the Kun–Thom criterion.

- There is one fixed finitely presented non-MF group with a computable family of words  $w_p$ , each invisible to every operator norm matrix model and lying in the commutator of the MF radical with the group, such that  $w_p = 1$  exactly when the program  $p$  halts.
- There is a finitely presented, point-norm matricially stable, non-MF group whose MF radical is the normal closure of a single nontrivial element.

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